



Published in final edited form as:

Cancer Discov. 2024 September 04; 14(9): 1631–1652. doi:10.1158/2159-8290.CD-23-0798.

Pre-existing skin-resident CD8 and $\gamma\delta$ T cell circuits mediate immune response in Merkel cell carcinoma and predict immunotherapy efficacy

Zachary Z. Reinstein^{1,2,3,4}, Yue Zhang^{1,4}, Oscar E. Ospina⁵, Matt D. Nichols⁶, Victoria A. Chu^{1,4}, Alvaro de Mingo Pulido⁶, Karol Prieto⁶, Jonathan V. Nguyen⁷, Rui Yin^{8,9}, Carlos Moran Segura⁷, Ahmed Usman^{1,4}, Brittney Sell⁶, Spencer Ng^{1,3,4}, Janis V. de la Iglesia⁷, Sunandana Chandra⁴, Jeffrey A. Sosman^{1,4,10}, Raymond J. Cho¹¹, Jeffrey B. Cheng^{11,12}, Ellie Ivanova¹³, Sergei B. Koralov^{13,14}, Robbert J. C. Slebos¹⁵, Christine H. Chung¹⁵, Nikhil I. Khushalani^{16,17}, Jane L. Messina^{7,16,17}, Amod A. Sarnaik^{16,17}, Jonathan S. Zager^{16,17}, Vernon K. Sondak^{16,17}, Charles Vaske¹⁸, Sungjune Kim¹⁹, Andrew S. Brohl^{16,17}, Xinlei Mi²⁰, Brian Pierce^{8,9,21}, Xuefeng Wang⁵, Brooke L. Fridley⁵, Kenneth Y. Tsai^{6,7,17,*}, Jaehyuk Choi^{1,2,3,4,22,23,*}

¹Department of Dermatology, Northwestern University Feinberg School of Medicine, Chicago, IL, USA

²Department of Biochemistry and Molecular Genetics, Northwestern University Feinberg School of Medicine, Chicago, IL, USA

³Center for Human Immunobiology, Northwestern University Feinberg School of Medicine, Chicago, IL, USA

⁴Robert H. Lurie Comprehensive Cancer Center, Northwestern University Feinberg School of Medicine, Chicago, IL, USA

⁵Department of Biostatistics and Bioinformatics, H. Lee Moffitt Cancer Center and Research Institute, Tampa, FL, USA

⁶Department of Tumor Metastasis & Microenvironment, H. Lee Moffitt Cancer Center and Research Institute, Tampa, FL, USA

Corresponding authors: Jaehyuk Choi, Robert H. Lurie Comprehensive Cancer Center, Northwestern University, 303 E. Superior St. Room 5-115, Chicago, IL 60611, jaehyuk.choi@northwestern.edu, Kenneth Y. Tsai, H. Lee Moffitt Cancer Center & Research Institute, 12902 Magnolia Dr., SRB-3, Tampa, FL 33612, Kenneth.Tsai@moffitt.org.

*These authors jointly supervised the work

Authors' Contributions

Z.Z.R. designed and performed the experiments, analyzed the data, and wrote the manuscript. Y.Z., O.O., M.N., and B.L.F. organized the clinical metadata, analyzed the data, and wrote the manuscript. V.C. performed the experiments and analyzed the data. A. M. P., K. P., and B.S. designed and performed the scRNA-seq experiments at Moffitt. C.M.S. and J.N. performed the multiplex immunofluorescence staining and analysis. R.Y. and B.G.P. performed TCR structural modeling and analysis. A.U., S.C., and J.S. aided sample acquisition at Northwestern. S.N. performed GeoMx ROI selection and analyzed the data. R.J.C., J.B.C., E.N.I., and S. B. K. developed and assisted with the ECCITE-seq protocol. J. I., R.S., and C.H.C. performed immunohistochemical characterization and analysis. S.K., A.B., N.I.K., J.L.M., J.S.Z., V.K.S., and A.A.S. identified the subjects and procured the tissue at Moffitt. C. V. oversaw and performed RNA-seq on the initial cohort. X.M. and X. W. provided statistical analyses. K. Y. T. and J.C. conceived and supervised the study, analyzed, and interpreted the data, and wrote the manuscript.

Conflict of interest statement:

C.H.C. has received honoraria from Merck and Regeneron for serving in ad hoc Scientific Advisory Board, although this board is not directly involved in MCC immunotherapy. J.C. is a co-founder and stockholder of Moonlight Bio. J.C. is an inventor on a patent for enhancing adoptive T cell therapeutics; 63/412,300. The authors declare no other competing interests.

- ⁷Department of Pathology, H. Lee Moffitt Cancer Center and Research Institute, Tampa, FL, USA
- ⁸University of Maryland Institute for Bioscience and Biotechnology Research, Rockville, MD, USA
- ⁹Department of Cell Biology and Molecular Genetics, University of Maryland, College Park, MD, USA
- ¹⁰Department of Internal Medicine, Northwestern University Feinberg School of Medicine, Chicago, IL, USA
- ¹¹Department of Dermatology, University of California, San Francisco, San Francisco, CA, USA
- ¹²Department of Dermatology, Veterans Affairs Medical Center, San Francisco, CA, USA
- ¹³Department of Pathology, New York University School of Medicine, New York, NY, USA
- ¹⁴Kimmel Center for Biology and Medicine at the Skirball Institute, New York University, Grossman School of Medicine, New York, NY, USA
- ¹⁵Department of Head & Neck Oncology, H. Lee Moffitt Cancer Center and Research Institute, Tampa, FL, USA
- ¹⁶Department of Cutaneous Oncology, H. Lee Moffitt Cancer Center and Research Institute, Tampa, FL, USA
- ¹⁷Melanoma & Skin Cancer Center of Excellence, H. Lee Moffitt Cancer Center and Research Institute, Tampa, FL, USA
- ¹⁸Clear Labs, Santa Cruz, CA, USA
- ¹⁹Department of Radiation Oncology, H. Lee Moffitt Cancer Center and Research Institute, Tampa, FL, USA
- ²⁰Department of Preventive Medicine - Biostatistics Quantitative Data Sciences Core, Northwestern University, Chicago, IL, USA
- ²¹University of Maryland Marlene and Stewart Greenebaum Comprehensive Cancer Center, Baltimore, MD, USA
- ²²Center for Genetic Medicine, Northwestern University, Feinberg School of Medicine, Chicago, IL, USA
- ²³Center for Synthetic Biology, Northwestern University, Evanston, IL, USA

Abstract

Merkel cell carcinoma (MCC) is an aggressive neuroendocrine skin cancer with a ~50% response rate to immune checkpoint blockade (ICB) therapy. To identify predictive biomarkers, we integrated bulk and single-cell RNA-seq with spatial transcriptomics from a cohort of 186 samples from 116 patients, including bulk RNA-seq from 14 matched pairs pre- and post-ICB. In non-responders, tumors show evidence of increased tumor proliferation, neuronal stem cell markers, and IL-1. Responders have increased type I/II interferons and pre-existing tissue resident (Trm) CD8 or V δ 1 $\gamma\delta$ T cells that functionally converge with overlapping antigen-specific transcriptional programs and clonal expansion of public TCRs. Spatial transcriptomics demonstrated co-localization of T cells with B and dendritic cells, which supply chemokines and

co-stimulation. Lastly, ICB significantly increased clonal expansion or recruitment of Trm and Vδ1 cells in tumors specifically in responders, underscoring their therapeutic importance. These data identify potential clinically actionable biomarkers and therapeutic targets for MCC.

Introduction

Merkel cell carcinoma (MCC) is a highly aggressive primary neuroendocrine skin cancer, with high rates of metastasis and recurrence(1). There are two subtypes of MCC, with different etiologies and phenotypes. In 60–80% of MCCs, the Merkel cell polyomavirus (MCPyV) is mutated and clonally integrated(2). They express obligate viral oncoproteins, small and truncated large T antigens, which drive carcinogenesis in part by inactivating p53 and pRB(3). Whereas virus positive (VP) MCCs inactivate pRB and p53 through the expression of the small and large T oncoproteins, virus negative (VN) MCCs achieve the same molecular endpoint through near universal somatic mutation of both *RB1* and *TP53*(2,4).

VN-MCCs are immunogenic because of the high number of tumor neoantigens from high mutation burdens. VP-MCCs have a low mutation burden but are immunogenic because they constitutively express viral oncoproteins(4). Shared antigens in VP-MCCs ease identification of tumor-specific T cells across patients(5).

For metastatic disease, immunotherapy is the only Food and Drug Administration-approved therapy that significantly prolongs survival. Unfortunately, many patients have either primary (~50%) or acquired resistance (up to 25%) to immune checkpoint blockade (ICB)(6,7). To elucidate potential mechanisms, we compiled the largest MCC cohort with comprehensive multi-modal molecular analysis (n=116 patients), specifically focusing on the tumor microenvironment with bulk RNA-seq (n=71 patients), single-cell RNA-seq (n=360,069 cells from 64 patients), multiplexed immunofluorescence (n=41 patients), and spatial transcriptomics (n=20 patients). 46 patients had more than one sample. These methods complement each other, addressing limitations and offering insights across tumor-intrinsic and extrinsic factors. Our multi-omic studies identified clinically actionable biomarkers and therapeutic avenues for MCCs and potentially for other cancers.

Results

Compilation of a clinically and molecularly annotated MCC cohort

We collected the largest multi-institutional MCC dataset with multi-modal molecular characterization of 186 samples across 116 patients including 92 patients with evaluable ICB responses (Table 1, Fig. 1A, Supplementary Tables S1–S2, Methods). These include bulk RNA-Seq (n= 85 samples from 71 patients) (Supplementary Tables S3–S5, Supplementary Fig. S1A) and a modified form of ECCITE-Seq (n= 54 tumor samples from 48 patients; n= 55 blood samples from 32 patients) (Supplementary Table S6–S7, Supplementary Fig. S1B), including three samples previously reported(8). Moreover, 44 distinct samples were spatially profiled using multiplex immunofluorescence (mIF) imaging (Akoya Phenoimager), spatial cancer gene or whole transcriptome profiling (Nanostring GeoMx), and single-cell

resolution spatial transcriptomics (Nanostring CosMx) (Fig. 1B, Supplementary Tables S8–S10, Supplementary Fig. S1C). Overall, 63 patients in our dataset were profiled using two or more modalities (Supplementary Fig. S2).

Bulk RNA-seq reveals predictive biomarkers of response to ICB

First, we analyzed the bulk RNA-seq of tumors, pre-ICB, from a cohort of 63 patients with 62 patients that were systemic treatment naive to minimize confounding variables related to prior therapy. Within this dataset, there were 31 responders (partial or complete response) and 32 non-responders (stable and/or progressive disease) (Supplementary Fig. S1A, Supplementary Table S3, Methods).

Differential gene expression analysis revealed 800 upregulated genes in responders and 436 in non-responders ($p_{adj} < 0.1$, DESeq2.0) (Fig. 1C, Supplementary Table S11). Many of the genes are characteristic of CD8 T cells ($\alpha\beta$ TCR, *GZMA*, *CD8A*, *CD8B*), $\gamma\delta$ T cells ($\gamma\delta$ TCRs), B cells (IG genes, *CD79A*), and plasmacytoid dendritic cells (*TLR7*, *LILRB4*). Several molecules were associated with tumor-specific cytotoxic T cells, including markers of T cell exhaustion (*HAVCR2*, *LAG3*, and *TIGIT*) and cytotoxic activity (*GZMB* and *PRF1*). Collectively, these data suggest an enrichment of pre-existing CD8 T cells, B cells, and specific myeloid subtypes in responders.

Pathway analysis with Enrichr(9) showed upregulation of IFN- γ signaling ($P_{adj} = 2.22e-63$), IFN- α signaling ($P_{adj} = 8.53e-52$), and antigen presentation machinery ($P_{adj} = 2.36e-17$), associated with IFN- γ signaling in other cancers including melanoma(10) (Fig. 1D, 1E, Supplementary Table S12).

Non-responders had upregulated MCC-specific genes, like *PAX6* and *FOXA1*. (Supplementary Table S11). Pathway analysis revealed additional enrichment of genes associated: 1) IL-1 processing(11), including *IL1A* and *IL1B*, and 2) E2F targets, which are normally expressed during cell division (Fig. 1F, 1G, Supplementary Table S12).

Single cell RNA-seq reveals cellular composition of the tumor microenvironment and blood

Next, we conducted single-cell RNA sequencing (scRNA-seq) initially on 49 tumor samples derived from 43 patients (Supplementary Table S6, Supplementary Fig. S1B). This included 8 samples from 4 patients across two timepoints (including 2 pairs of samples collected before and after ICB), 4 samples from 2 patients spanning different tissues (skin/lymph node pairs), and 37 single samples. Additionally, we analyzed 55 peripheral blood mononuclear cell (PBMC) samples from 32 patients. This included 21 pre/post-ICB pairs (42 samples), 2 samples from a third timepoint, and 11 samples that were unmatched. 26 of these samples were matched to patients that also had scRNA-seq on tumor tissue. (Fig. 1A, 1B, Supplementary Fig. S3–5, Supplementary Table S7). 313,757 cells were recovered from these samples using scRNA-seq. An additional cohort of 5 post treatment tumor samples with 46,312 cells were integrated and analyzed separately to investigate the cellular landscape after ICB for a total of 360,069 cells. Following quality control, we identified 27 distinct cell types in the tumor immune microenvironment, including CD8 T cells, CD4 T cells, natural killer (NK) cells, $\gamma\delta$ T cells, macrophages, dendritic cells, plasma cells, B

cells, and plasmacytoid dendritic cell (pDC) clusters (Fig. 2A, 2B, Supplementary Fig. S3, S4A, Supplementary Table S13).

Types I and II Interferon signatures in tumor cells predict response

First, we analyzed the 60,902 tumor cells, which were marked by the expression of neuroendocrine transcription factor *ATOH1* and chromogranin(3). We examined expression of the viral oncogenes small T antigen (sTA_g) or large T antigen (LTA_g) by scRNA-seq and bulk RNA-seq. 26/43 patients had samples with multiple reads for both sTA_g and LTA_g and were thus considered VP. The remainder had no transcriptional evidence of MCPyV sTA_g or LTA_g expression, which we labeled VN.

Next, we performed differential gene and pathway expression analyses (Supplementary Table S14, Fig. 2C). In responders, we observed upregulation of multiple pathways. Similar to bulk RNA-seq, there was upregulation of genes associated with IFN- γ (*STAT1*, *MX1*, *IRF1*) as well as IFN- α (*OAS1*, *IFI27*, *IRF9*). Again, many IFN- γ -associated genes are implicated in antigen presentation (i.e., *HLA-I*, *HLA-II*, *ERAP2*) (Fig. 2D, Supplementary Table S15). Next, we ran SCENIC regulon analysis to identify significant transcription factor-associated transcriptional networks. Top regulons in responders highlighted type I/II interferon responses (IRF7, IRF9, STAT1, IRF1). (Fig. 2E, Supplementary Table S16).

We calculated a module score to quantify IFN- γ and IFN responses using previously described methods(10,12). IFN- α and IFN- γ response module scores were highly correlated (Pearson correlation, $R^2 = 0.928$, $p < 1e-10$). Both signatures were associated with the infiltration of immune cells, suggestive of pre-existing immunity (Fig. 2F). The source of type I interferons was not clear because it was not well captured by scRNA-seq. However, subsequent analyses showed that the IFN- α module score in multiple cell types (tumor cells, T/NK cells, and stroma) was correlated with the presence of pDCs, which express high levels of type I interferons(13) (Pearson correlation, $R = 0.38$, $p = 0.011$ in tumor cells) (Supplementary Fig. S6A–B). Other possible candidate cell types showed lower correlations (classical dendritic cell (cDC) $R = 0.045$, macrophage $R = -0.0010$). The source of IFN- γ appeared to be primarily CD8 and other NK/T-cells (Fig. 2G, Supplementary Fig. S7A). The enrichment of type I/II interferon gene sets in responders did not depend on the site of tissue sampling (LN versus skin) or metastatic status (primary versus metastasis) (Supplementary Fig. S7B).

Tumor cell proliferation is anti-correlated with immune cell infiltration and survival post ICB

In non-responders, we found upregulation of 21 genes and 8 Hallmark pathways via GSEA. These include MYC (Padj = $3.05e-17$), mTORC1 (Padj = $1.68e-10$), and E2F (Padj = $7.62e-4$). We hypothesized that the E2F signatures observed in bulk RNA-seq arose from tumor cells. To test this hypothesis, we first identified cycling tumor cells. These cells expressed markers of cell cycle/division, *MKI67*, *TOP2A*, *CENPF* and E2F targets, such as *CCNB2* and *CDKN3*, varying in proportion across samples from 0 to 5% of the total tumor cells. Because mTORC1 and MYC can be associated with proliferation, we confirmed that

they retained significance even when controlled for cell cycle (Supplementary Fig. S8A–B, Methods).

Consistent with data from other tumor types(14), we observed that proliferating tumor cells was negatively correlated with both overall immune infiltration (Pearson correlation, $R = -0.6$, $p=1e-5$) and the IFN- γ response, which we posit is a response to IFN- γ producing lymphocytes in the TME (Pearson correlation, $R = -0.6$, $p=4e-5$) (Fig. 2H). Finally, to test the effects on survival, we derived a tumor cycling signature and applied it to our bulk RNA-seq, which is significantly larger than our scRNA-seq dataset. Samples from the bulk RNA-seq dataset were scored based on the “tumor cycling” cell type signature derived from our single cell dataset with gene set variance analysis (GSVA) and split into the top or bottom quartile (see Methods). Patients with high levels of the tumor cycling signature had worse overall survival post ICB (Fig. 2I, Supplementary Table S17) ($p = 0.016$, Cox proportional hazard).

To find additional drivers of primary resistance, we analyzed transcription factors upregulated in tumors from non-responders. The most significantly upregulated gene in non-responders was OTX2, a transcription factor involved in neural cell development. SCENIC analysis(15) confirmed that OTX2 target genes are also upregulated (Fig. 2E). The analysis identified additional transcription factors associated with neural stem cells (HOXA10, GATA2, and SIX1) (Supplementary Table S16)(16–18).

Collectively, tumor cell analysis identified genes and signatures predictive of the response to ICB that correlate with the numbers of immune cells and effector molecules produced by them in the tumor microenvironment. To assess this further, we analyzed the immune infiltrates.

CD8 T cell analysis reveals two independent trajectories

We turned next to CD8 T cells because they express the highest levels of IFN- γ in the tumor microenvironment (Fig. 2G, Supplementary Fig. S7B). Single-cell analysis identified 15 distinct clusters that were organized into two distinct meta-clusters (Fig. 3A). The first, which we termed circulating T cells (Tcirc), included CD8 clusters C00_CD8, C01_CD8, C02_CD8, C09_CD8, C11_CD8, and C14_CD8. This cluster is enriched in T cells normally found in circulation, e.g. naïve, central memory, and terminally differentiated effector memory (Temra) T cells. Cells within this meta-cluster have upregulated genes or proteins associated with naïve and central memory cells. This includes markers of stemness, e.g. *IL7R*, *TCF7*, and *LTB*(19) markers of blood circulation, e.g. *CCR7*, *SELL*, *S1PR1*, and *KLF2*(19). Changes in a subset of genes was confirmed at the protein level including CD127, CD62L, and CD45RA, which are upregulated in naïve, central memory, and Temra cells (Supplementary Table S18, Fig. 3B–D).

The second meta-cluster (comprising CD8 clusters C03_CD8, C04_CD8, C05_CD8, C06_CD8, C07_CD8, C08_CD8, C10_CD8, C12_CD8, and C13_CD8) showed the upregulation of genes associated with tissue-resident T (Trm) differentiation. These include adhesion genes (*ITGAE*, *ITGA1*, *CD69*), chemokine receptors normally found in skin-

resident T cells (*CXCR6*), transcription factors (*PRDM1* and *RUNX3*)(20), and skin Trm-specific metabolic genes (*FABP5*)(21) (Fig. 3D, Supplementary Table S19).

Trm cells displayed significantly increased expression of genes reflecting recent activation (*41BB*, *CD38*), as well as markers of chronic activation/exhaustion (*LAG3*, *HAVCR2*, *PDCD1*, *ENTPD1*, *TOX*) (Fig. 3C) (X^2 , $p < 2.2e-16$). Based on the antibody derived tags (ADTs), the most common surface phenotype of Trm cells in MCCs was CD103+, CD137+, CD39+, CD279+, TIGIT+, CD8 T cells (Fig. 3B). With the addition of RNA markers, the majority of Trm cells also expressed *LAG3* and *HAVCR2* (Fig. 3C, 3D). Trm cells had over 2-fold increased expression of IFN- γ compared to Tcirc cells, with clusters C03_CD8, C05_CD8, C07_CD8, and C13_CD8 showing the highest levels (Supplementary Table S20, Supplementary Fig. S9). They also showed the highest expression of cytotoxic enzymes including *GZMB* and *PRF1* (Fig. 3B, 3D).

To elucidate the relationship between Trm and Tcirc meta-clusters, we performed two analyses. First, diffusion mapping revealed two distinct developmental lineages (Fig. 3E), which largely mirrored Trm and Tcirc meta-clusters. Second, we performed clonotype linkage analysis, which identifies CD8 clusters that have shared TCR clones and are thus developmentally linked. Hierarchical clustering highlighted the two Trm and Tcirc meta-clusters (Supplementary Table S21, Fig. 3F). Analysis of individual TCRs underscored the lack of inter-mixing, as 88% of expanded T cell clones were exclusively restricted to either the Trm or Tcirc meta-cluster (Fig. 3G), which is unlikely to occur by chance alone (Monte-Carlo simulation, $p < 1e6$) (Supplementary Fig. S10A–B).

To confirm that Tcirc cells were derived from the blood, we compared the clonotypic overlap in patient-matched scRNA-seq from the blood. Consistent with our expectation, we found that Tcirc cells shared 4.95-fold more clonotypes with CD8 T cells found in the blood than Trm cells ($p < 2.2e-16$, X^2) (Fig. 3H).

Pseudotime analysis of tissue CD8 T cells provided further insights into the molecular characteristics of these trajectories. We found that Tcirc cells differentiated from a naïve state to a circulating effector memory state, exhibiting decreased expression of naïve markers *IL7R* and *CCR7*, increased expression of *GZMH*, and a subsequent spike in *GZMK* expression, coinciding with *JUND* expression (Supplementary Fig. S11A, B). Trm cells acquired exhaustion markers during differentiation (*TOX*, *PDCD1*) and lost the expression of genes such as *BTG1*, which may function as a negative regulator of effector activity(22) (Supplementary Fig. S11C, D).

Trm cells are tumor-specific and are prognostic of ICB response

We hypothesized that Trm CD8 T cells would be enriched for tumor antigen-specific cells because they showed the highest expression of both activation/exhaustion markers (proteins 41BB, CD38, PD-1, and the *TOX* gene) and cytotoxic enzymes (*GZMA* and *GZMB*) (Fig. 3B–D). Moreover, the Trm lineage showed significantly higher levels of clonal expansion (Supplementary Table S22, Fig. 3I–J, Supplementary Fig. S12A–B).

To test this, we measured the NeoTCR8 score, a gene signature previously found to be enriched in tumor neoantigen-reactive T cells in multiple solid tumor types(23). The highest NeoTCR8 scores were found in the Trm clusters, with all but one of the Trm clusters ranked higher than any of the Tcirc clusters (Fig. 4A, Supplementary Fig. S9). Importantly, we found that this signature was similar across VP and VN-MCCs and was independent of site of tumor sampling or stage (Supplementary Fig. S12C–D, Supplementary Fig. S13A–C).

VP MCCs showed larger TCR clone sizes among CD8 T cells compared to VN MCC. This may be because of a restricted tumor antigen space due to uniform expression of immunogenic viral proteins and few, if any, private somatic mutations/neoantigens(4). This feature facilitated functional validation. We transduced Jurkat-76 cells ($\alpha\beta$ TCR knockout) with an NFAT-GFP reporter with four highly expanded TCRs (3 Trm-restricted, 1 Tcirc-restricted) and co-cultured them with autologous lymphoblastoid cell lines (LCLs) pulsed with peptide pools. These pools contain 116-amino acid peptides that tile along both the small T antigen and the large T antigen of MCPyV (Supplementary Table S23)(24). All Trm-restricted TCRs demonstrated antigen-specific activation in response to a unique peptide antigen, confirming viral specificity (Fig. 4B–C, Supplementary Fig. S14A–D). The expanded Tcirc TCR did not show any specific activation (Supplementary Fig. S14E). Structural modeling supported the antigen specificities of the TCRs tested. When the TCR sequences were modeled with the experimentally-validated peptides and the associated predicted MHC, 2/3 Trm TCRs had a TCRmodel2 score >0.8 out of 1, indicating high confidence of binding of the TCR to the functionally-validated peptide-MHC complex(25) (Fig. 4D, Supplementary Fig. S14F).

To link T cell receptors to public Merkel cell polyomavirus antigens within and across patients with VP-MCCs, we deployed GLIPH2, a computational pipeline that identifies biochemical similarities across complementarity-determining regions 3 (CDR3)(26). We found 10 GLIPH2 motifs among 23 unique TCRs that were highly restricted to the VP cohort (>85%) and were highly expanded (>20 clones), suggesting that they respond to tumor-specific Merkel cell polyomavirus antigens (Fig. 4E). These TCRs were also highly enriched in the Trm cells (Supplementary Table S24). In contrast, we found fewer GLIPH motifs (n=5) enriched in the VN cohort. Unlike for VP MCC, none of these motifs was expanded (i.e. clones of more than one cell) in multiple patients (Supplementary Table S24). This is consistent with the relative absence of public antigens in VN-MCCs where private mutations are created by UV-mediated mutagenesis.

To further explore the antigen specificity of MCC TILs, we compared our TCR sequences with known TCRs epitopes from three TCR databases that have compiled TCRs with known specificity to common viral pathogens that are not associated with MCC(27–29). We found 128 clonotypes that were predicted to bind to public viral antigens not expressed by Merkel cell carcinomas (including cytomegalovirus, influenza, and SARS-CoV-2, but not MCPyV). Consistent with the absence of these antigens in MCC tumor cells, these TCRs showed minimal expansion in the tumor microenvironment (Supplementary Table S25). When present, these clonotypes were significantly enriched in the Tcirc cell trajectory ($\chi^2=19.4$, $p = .00001$) (Supplementary Fig. S15A).

Finally, we assessed the clinical significance of Trm cells. First, the prevalence of Trm cells was correlated with response to ICB ($p=0.018$) (Supplementary Fig. S15B). This was corroborated by multiplex IF data performed on 44 samples from 38 patients (24 with assessable response), of which 17 had associated tumor scRNA-seq. No one marker set defines Trm cells; however, our single cell data suggest that CD103 is expressed in 38.6%, a plurality of Trm cells. CD3+CD8+CD103+ skin-resident T cells were observed infiltrating the tumor parenchyma (Fig. 4F), and tumors from responders had significantly higher densities of CD3+CD8+CD103+ cells as compared to non-responders in our mIF panel (Fig.4).

Strikingly, the ratio of Trm to Tcirm cells was also predictive of clinical outcomes. The ratio of Trm:Tcirm was higher in responders, with all non-responders having equal or more Tcirm cells compared to Trm cells. (Fig. 4H). This was confirmed in the bulk RNA-seq using GSVA on a Trm signature we derived from our single cell dataset (see Methods). Patients with tumors with high expression of the Trm signature (the top quartile) had significantly improved median survival from the start of ICB compared to those with low Trm signatures (the bottom quartile) (median survival not reached vs. 578 days, respectively, $p = 0.038$, Cox proportional hazard) (Fig. 4I). Collectively, these data show that the presence of Trm cells is associated with both response to ICB and longer survival.

Vδ1 $\gamma\delta$ T cells are enriched in tumors of responders with low IFN- γ signatures

Next, we focused on samples from patients who responded to immunotherapy despite having a low IFN- γ signature (Fig. 5A). This analysis identified non-conventional cytotoxic T cells as potential prognostic biomarkers. Single-cell analysis identified 17 clusters of $\gamma\delta$ and NK cells (Fig. 5B). To unequivocally identify $\gamma\delta$ T cells, we utilized TCR data obtained through our modification of ECCITE-seq(30), which enriches $\gamma\delta$ TCR sequences. With one exception, most patients had low levels of NK cells (<2% of cells). In contrast, we found that $\gamma\delta$ T cells were enriched (5.7–25.0% of all cells) in 7/49 of tumor samples, including four samples from responders with a low IFN- γ signature (Fig. 5A). We observed that tumors with low MHC-I expression were enriched for expression of *TRDV1*, which encodes Vδ1 TCR (Fig. 5C).

$\gamma\delta$ T cells have three developmentally distinct cell types distinguished by the choice of Vδ segment during VDJ recombination(31). Using our $\gamma\delta$ -specific TCR enrichment protocol, we found that Vδ1 cells in MCCs preferentially paired with V γ 3 and V γ 4 chains, and Vδ2 cells predominantly paired with the V γ 9 chain. We identified nine clusters of $\gamma\delta$ cells: seven Vδ1 and two Vδ2, with the few Vδ3 cells primarily found in the Vδ1 clusters (Fig. 5B, 5D).

The predominant $\gamma\delta$ cells among the MCC TILs were Vδ1 cells, consistent with previous reports(32) (Fig. 5D). Similar to CD8 Trm cells in MCC, Vδ1 and Vδ3 cells are tissue resident. In contrast, Vδ2 cells are the primary $\gamma\delta$ cell in the circulation. Vδ1 and Vδ3 cells (but not Vδ2 cells) appeared to undergo antigen-dependent activation. Vδ1 cells expressed multiple markers of co-inhibition (*LAG3*, *PDCD1*, *TIM3*), activation (*41BB*, *KLRC2*), and effector functions (*GZMA*, *GZMB*, *FASLG*), as well as multiple inhibitory KIRs (*KIR2DL1*, *KIR2DL3*), and transcription factors associated with CD8 exhaustion (*EOMES*, *BATF*) or tissue residence (*RUNX3*). Many of these were significantly enriched compared

to those in Vδ2 cells (Supplementary Tables S26–S28, Fig. 5E–G). Conversely, Vδ2 cells showed high expression of circulatory markers (*CCR7*, *CD127*, *CD62L*) along with higher expression of AP-1 family members, thus broadly matching the expression patterns observed in CD8 Tcirc cells (Fig. 3D).

Using CITE-seq antibody-based analysis, tumor-resident Vδ1 cells expressed less surface CD3 compared to circulating Vδ1 cells and/or Vδ2 cells in the blood or tumor, reflective of receptor downregulation following activation (Supplementary Table S28, Fig. 5F). For Vδ1 cells, there were 397 unique clonotypes, of which 128 (32.2%) were expanded, suggesting higher clonal proliferation and clonal focusing than in Vδ2 cells (372 total clones, 85 expanded). The median clonal size was significantly higher among Vδ1 (17 vs. 3, $p < 2.2 \times 10^{-16}$, Wilcoxon test) (Supplementary Fig. S16A). Almost all the expanded TCRs from 128 clones had biochemically similar δ -chain CDR3s (see Methods), suggesting that they responded in a clonotype-specific manner to a public antigen, whereas the rest of the Vδ1 CDR3s were not expanded and were not similar, suggesting a diverse background of TCRs that may not recognize the same antigen (Fig. 5H).

Pseudotime analyses suggest that Vδ1 cells start from a naïve-like population (cluster C08_GD), which consisted of T cell clones that did not proliferate (as determined by clonotype analysis) and enriched for the expression of markers associated with naïve $\gamma\delta$ T cells, including *CD27*, *CD28*, *CD45RA*, *CD127*, *CCR7* and *TCF7*(33) (Supplementary Fig. S16B). The Vδ1 trajectory shifted primarily toward clusters C02_GD, C15_GD, and C10_GD, matching the clonal expansion observed previously (Fig. 5I, Supplementary Fig. S16C). These differentiated cells had higher levels of cell cycle markers (*MKI67*), exhaustion markers (*ENTPD1*, *LAG3*, *HAVCR2*, *TIGIT*, *BATF*), cytotoxic effectors (*GZMH* and *GZMB*), and cell migration proteins (*CXCR6* and *VCAM1*) (Supplementary Table S29, Fig. 5J). These data suggest that Vδ1 cells are CD8 T cell-like, in that they differentiate to a chronically activated phenotype in response to a CDR3-dependent antigen (Fig. 3D). Unfortunately, there were insufficient cells to perform pseudotime analysis of Vδ2 or Vδ3 cells.

To orthogonally validate the functional relevance of $\gamma\delta$ T cells in MCCs, we performed multiplex IF on seven patients (eight samples), including four with RNA-seq signatures suggestive of $\gamma\delta$ T cell infiltration. For a subset of MCCs, $\gamma\delta$ cells were also detected in the tumor parenchyma expressing cytotoxic markers (Granzyme B) (Fig. 5K). Interestingly, we found no statistically significant differences in Vδ1 infiltration between skin and LN samples in either our single cell data or bulk RNA-seq data. All highly enriched Vδ1 samples were found in the skin, but the numbers of samples preclude statistical significance (Supplementary Fig. S13).

Spatial transcriptomics localizes key immune signatures and a cellular triad associated with response

To resolve the TME spatially, we utilized first used the GeoMx Digital Spatial Profiler, we identified 180 Regions of Interest (ROIs) across 15 samples from 12 patients. These 12 patients included four complete responders to ICB after biopsy and eight non-

responders. The varied ROIs allowed for a comprehensive overview of the tissue landscape (Supplementary Fig. S17A–B, S18A).

Differential expression analysis (DE-seq) was performed on these ROIs. Within the tumor region of complete responders (CR), we observed both T cell genes (*LAG3*, *GZMB*, *TRDC*) and interferon response genes (*GBP4*), while in non-responders (all with progressive disease), there was prominent expression of tumor markers (keratins, *CD44*, *HOX* genes) and proliferation markers (*TYMS*, *MKI67*) (Supplementary Table S30, Supplementary Fig. S18B).

We detected a strong interferon-gamma (IFN- γ) response signature (*GBP4*, HLA-I, TAP, and proteasome genes) in the stroma of CR samples. Additionally, T cell markers (*CD27* and *CD7*), B cell markers (*CD79A* and *POU2AF1*), and macrophage markers (*C1QB*, *CD68*, *CD163*) were identified, confirming the localization of these cells in the stroma. Conversely, in progressive disease samples, we identified tumor markers (keratins, *PRAME*) and proliferation markers (*TYMS*, *MCM*, *E2F1*) (Supplementary Table S31, Supplementary Fig. S18B). We utilized the Trm and Tc1r gene signatures that we had already developed to spatially examine these specific T cell lineages. We observed that while both Trm and Tc1r were upregulated in the stroma of responders, only Trm was upregulated in the tumors of responders. Non-responders failed to show enrichment of Trm or Tc1r genes, regardless of location (Fig. 6A).

Next, for single cell resolution, we performed CosMx Spatial Molecular Imager analysis. We selected 122 fields of view (FOV) sized 0.9 mm x 0.7 mm across the tumor-stromal interface from 13 samples across 12 patients (6 responders, 4 non-responders). We segmented 614,209 cells (Supplementary Table S32, Fig. 6B, Supplementary Fig. S19A–D, Methods). Next, we deployed SpatialTIME, a permutation-based method that we developed to generate a degree of clustering (DoC) score to rank whether cell populations are more co-localized vs. more dispersed than a distribution based on complete spatial randomness across all possible binary comparisons at specified neighborhood radii(34). We then used the presence of elevated DoC to identify key cellular immune circuits associated with ICB response. To do this, we performed unsupervised hierarchical clustering to identify four cell types (T cells, B cells, pDC, and DC) exhibiting high DoC across all FOVs and the entire range of radii $r = 250\text{--}500$ (90 – 180 μm diameter) (Supplementary Table S33, Fig. 6C, Supplementary Fig. S20A–D). The highest DoC corresponding to the highest levels of co-localization ($> 4.5 \times 10^5$) was seen across two circuits: T cell – B cell – DC and T cell – B cell – pDC. An example of this is seen in an FOV highlighting T cells, B cells, and pDC all co-localized at the tumor-stroma interface (Fig. 6D). We further sought to identify whether these distributions differed between responders and non-responders. Among responders, B cells, DCs, and pDCs showed 61.9-, 1.8-, and 14.3-fold higher counts, respectively, with a $r=300$ (108 μm diameter) in the neighborhood around T cells ($p < 2.2\text{e-}16$, χ^2) (Fig. 6E, Supplementary Fig. S20E). Conversely, T cells were preferentially co-localized with 1.7- and 13.5-fold more tumor and tumor cycling cells, respectively, in the same sized neighborhoods ($r=300$; 108 μm diameter), suggesting a paucity of intact immune circuits in non-responders (Fig. 6E, Supplementary Fig. S20E). We interpret this to suggest that the

presence of a key immunological triad of B cells, T cells, and DCs or pDCs is correlated with ICB response.

We then leveraged the droplet-based scRNA-seq dataset to identify receptor-ligand interactions because of higher numbers of transcripts per gene, focusing on B cell and myeloid populations (including dendritic cell subsets) (Supplementary Fig. S6A). Using CellChat(35), we found that B cells, DCs, and pDCs interacted with Trm T cells via multiple chemokine-chemokine receptor (i.e. *CXCL16* to *CXCR6*), costimulatory ligand-receptor interactions (i.e. 41BBL (*TNFSF9*) to 41BB (*TNFRSF9*), *CD80/CD86* to *CD28*), and activating cytokines (i.e. *IL15* to IL-15 receptor, *IFNA2/IFNW1* to *IFNAR1/2* (IFN- α receptor))(Supplementary Table S34, Fig. 6F).

ICB increases recruitment of Trm cells and multiple inflammatory pathways in responders

To examine the effects of ICB, we examined a cohort of 7 responders and 7 non-responders from which we sampled matched tumor tissue pre- and post-therapy. Among responders, 1560 genes were upregulated and 995 downregulated after therapy. (DE-Seq2, $P_{adj.} < 0.1$) (Supplementary Table S35). In contrast, only two differentially expressed genes were identified among non-responders (Fig. 6G).

Overall, responders showed upregulation (1.34–1.44 fold, $p < 0.05$) of multiple inflammatory signatures after treatment including our V δ 1 signature, and an IFNG signature(10). There was a strong overlap in the pathways and the genes upregulated in pre-ICB responders and those upregulated in responders post-ICB (Fig. 6H, Supplementary Table S36, Supplementary Fig. S21A, B). These include markers of B cells (e.g. immunoglobulins), pDC/DC markers (e.g. *PTGDS*), T cell markers (e.g. CD3, TCR), chemokines/cytokines (e.g. *IFNG*, *IL15*, *CXCL16*), CD8 Trm markers (e.g. *CD69*, *PRF1*, *CD8B*, *CXCR6*), and multiple interferon response genes (e.g. HLA, *IRF1*, *IFI16*) and gene signatures of Trm cells (but not Tcirc cells)(1.44-fold, $p=0.031$; Wilcoxon; Supplementary Fig. S21B). These data collectively suggest that ICB increases the recruitment or expansion of immune cells that were present prior to ICB. Cell cycle and cell proliferation pathways were downregulated in responders post treatment. These included markers of tumor proliferation (*FOXM1*, *TOP2A*, *MKI67*, *E2F7*) and MCC specific markers (e.g. *ATOH1*, *NEFM*), suggesting tumor clearance. In contrast, non-responders show little to no change after ICB treatment (2–9% change, $p > 0.5$) and were thus lower than temporally matched responders pre- or post-ICB (Supplementary Fig. S21).

Next, we examined a cohort of scRNA-seq samples only acquired from patients after immunotherapy. These 15 samples included 5 additional samples that were projected on the original single cell dataset (see Methods). Consistent with our bulk RNA-seq data, responders ($n=5$) had 11.1-fold lower levels of tumor cycling cells and 3.01-fold higher CD8 Trm cells (3.01-fold increase) following ICB ($p < 0.05$, Wilcoxon) (Fig. 6I, Supplementary Fig. S21C).

Next, we examined antigen receptor sequences to study clonal dynamics in the TME after ICB in our bulk RNA-seq data using MIXCR(36). As suggested in our differential expression analysis, we observed 60 to 400-fold more total TCR or BCR clones in

responders versus non-responders (Wilcoxon, $p < 0.05$). Non-responders had no expanded TCRs and BCRs prior to treatment, while 3/7 responders had expanded TCR clones and 5/7 responders had expanded BCR clones prior to treatment. After treatment, 1/7 (TCR) and 2/7 (BCR) non-responder samples showed evidence of new, expanded clones. In contrast, all of the responders had expansion of newly-recruited and pre-existing TCR and BCR clones (Supplementary Fig. S22A–D, see Methods). Thus, both pre-existing immunity and the recruitment of new, reactive T-cells and B cells are implicated in response.

Since MIXCR data can be biased by differences in TCR mRNA between cells, we used our matched single cell data to more precisely assess CD8 T cell dynamics. We examined two paired sets of ICB responder samples with pre- and post-therapy scRNA-seq from both the blood and tissue. One sample pair had matched bulk RNA-seq and one pair was new. After treatment, we saw the same results as our MIXCR data, where both samples had expansion of pre-existing clones, but also recruitment of expanded clones (Supplementary Fig. S23A–B). Expanded cells from both patients were enriched for putative Trm-like signatures. They expressed CD103 and exhaustion markers (exhaustion score(37), PD-1, and *LAG3*). Following anti-PD1 therapy, expanded cells had significantly lower levels of exhaustion markers, including a reduced exhaustion score and a 2.11 to 5.44-fold decrease in PD-1 and *LAG3* expression, respectively ($p < 1e6$, Wilcoxon) (Fig. 6J, Supplementary Fig. S23C), which may reflect increased functionality after ICB. Full cell type analysis of pre-post matched scRNA-seq from the blood (21 pairs) did not reveal any significant cell clusters correlated with response or resistance. Tumor-specific T cells comprised a maximum of 0.5% of circulating T cells, but the number of cells per sample was not high enough to reliably capture a cluster of those cells(38).

Discussion

In this study, we compiled the largest database to date of MCC tumor samples and matched blood with multi-modal profiling of a total of 116 patients. The breadth and depth of our studies have allowed us to confirm findings across multiple orthogonal datasets, highlight the importance of tumor sampling (over blood), and identify multiple novel candidate biomarkers that may be clinically actionable. For instance, the negative impact of tumor cell proliferation on ICB response (Fig. 1F–G, 2H–I) may be validated in CLIA settings using assays assessing Ki-67 expression. Moreover, next-generation sequencing may similarly enable prospective validation of IFN- γ and/or IFN- α signatures to predict responses to immunotherapy(10) (Fig. 1D–E, 2D,F).

Our data emphasize the role of tissue-resident lymphocytes in MCCs. Despite representing a minority (~45%) of the total CD8 T cells, tissue-resident CD8 T cells appear to be the primary tumor-specific population responsible for tumor recognition and killing. Moreover, the prevalence of this tissue-resident population (but not the circulating T cell population) is clinically important. In contrast to some other cancer types(39), their presence in the tumor prior to immunotherapy predicted responses (Fig. 4H–I).

Our study provides novel insights into $\gamma\delta$ T cells in the TME. While others have reported significant infiltrates of $\gamma\delta$ T cells(32), our modified ECCITE-seq allowed unique $\gamma\delta$

TCR sequencing with protein markers, which enables linkage of VDJ, transcriptome, and proteomics. This analysis clearly indicated that skin-resident V γ 3V δ 1 and V γ 4V δ 1 cells were the tumor-reactive population (and not the circulating V γ 9V δ 2 cells) (Fig. 5E–J).

V δ 1 cells in MCCs expressed costimulatory and inhibitory receptors commonly expressed by innate NK cells including killer cell lectin-like receptor (KLR)/KIR family members (Fig. 5E, J). However, our analysis highlighted deep commonalities with CD8 Trm cells. Like CD8 Trm cells, they express acute activation markers (such as 41BB), undergo clonal expansion, and acquire markers associated with T cell exhaustion (e.g., PD-1, *LAG3*, TIGIT). The pseudotime analyses paralleled those of CD8 Trm cells, wherein they both started in a *TCF7*⁺ cell state and ended in an exhaustion state and loss of *BTG1*, a transcriptional repressor of lymphocyte effector function (Fig. 5J)(22). Taken together, these skin-resident V δ 1 cells appear to display a form of functional convergence with skin-resident CD8 Trm cells in driving anti-tumor immunity (Supplementary Fig. S24A–B).

Consistent with recent reports, these V δ 1 cells are enriched in tumors with altered MHC-I expression(40). In contrast to previous reports, our TCR sequencing data suggest that the CDR3 region is critically important for V δ 1 cells (Fig. 4D, 5I)(40,41). Through single-cell $\gamma\delta$ CDR3 sequencing, we demonstrated clonal expansion of V δ 1 cells that express biochemically similar CDR3s, highlighting the possibility of shared reactivity to a common (as yet unidentified) antigen (Fig. 5I).

Importantly, multiple spatial approaches suggest that T cells in the MCC TME do not function independently. Our data highlights the importance of cellular neighborhoods in response to immunotherapy (Fig. 6A–E).

Finally, we identified novel biomarkers, such as IL-1, type I interferon and neural stem cell markers. Both IL-1 and type I interferon have been described as having opposing pro- and anti-tumor roles in cancer(42–44). However, our data suggest that exogenous type I interferon and/or IL-1 blockade may enhance immunotherapy efficacy. OTX2 and other neural stem cell markers may represent novel targets to improve response to ICB in MCC (Supplementary Fig. S25). MCCs may have unique, yet undiscovered, properties that highlight the biological importance of these molecules, specifically for this cancer type. Alternatively, this unique genomically homogeneous dataset (particularly VP-MCCs) may provide statistical power to make discoveries that are not possible for genetically heterogeneous cancers of other types. In other cancer types, these signals may be subsumed, for example, by the tumor mutation burden, which is required to make the tumor immunogenic. Thus, we believe that many of these results are applicable to other cancers, especially skin cancers.

Methods

Patient selection and response assessment:

A total of 116 MCC patients treated at three institutions (Moffitt Cancer Center, Northwestern University Lurie Cancer Center, University of Washington) were included for analysis. The cohort was 79% male, median age 73 years (range 18–93 years).

Disease stage and immunocompromised status was reported in Tables 1 and S1. Inclusion criteria were histologically confirmed diagnoses of Merkel cell carcinoma made within the Department of Pathology at the corresponding institution and treatment with systemic immunotherapy. The clinical response to immune checkpoint blockade was defined as complete response, partial response, stable disease or progressive disease. However for the purposes of all molecular analyses, characterization of clinical response to ICB was dichotomized to responder (complete and partial response) versus non-responder (stable and progressive disease) status. These assessments were made on manual review of all electronic medical records on the basis of a combination of histopathologic evaluation (for neoadjuvant therapy), clinical assessment (clinical assessment of skin and / or lymphadenopathy), serial radiographic assessment (computed tomography or positron emission tomography - computed tomography).

Complete responses were defined as complete radiographic and clinical response, or, where applicable, concordant pathological complete response. In the case of adjuvant immunotherapy, complete responses were noted in situations where surgery with curative intent had evaluable clinical, or radiographic residual disease burden which was subsequently treated with ICB with clearance of disease. Any intermediate responses with residual disease burden were considered partial responses.

Progressive disease was defined as enlargement of overall burden of disease as noted by a combination of clinical and radiological assessment and / or the emergence of new metastases. In the case of adjuvant ICB administered following surgery with or without radiation, any recurrence or progression was considered progressive disease (n=3). One patient on adjuvant ICB had stable disease with neither disease resolution nor progression. One mixed response was noted and considered an overall partial response in terms of overall tumor burden but the lesion used for molecular analysis showed no evidence of response to therapy and was considered as a progressive disease (MO_MCC_057). Death or date of last follow up was the clinical endpoint used for survival analysis.

Sample collection:

Fresh tissue and blood specimens were collected from patients with Merkel cell carcinoma (MCC) treated and managed at the H. Lee Moffitt Cancer Center Department of Cutaneous Oncology following written informed consent under the IRB-approved Total Cancer Care Protocol (MCC 14690, 94962) and performed in accordance with the Declaration of Helsinki. Collected specimens were processed and used under IRB-approved MCC Protocol 19191. Freshly collected tumor specimens were dissected for single cell dissociation, formalin fixation with subsequent paraffin embedding, or flash frozen. Additional archival tissues at Moffitt were processed under waiver of consent (MCC 19191).

At Northwestern, fresh samples were provided by the Skin Biology & Diseases Resource-Based Center (SBDRC) Translational & Experimental Skin Testing & Immune Tracing (TEST-IT) Core and were obtained through the IRB-approved Dermatology Tissue Acquisition and Repository (IRB # STU9443) or through the IRB-approved Dermatopathology Tissue Repository for Research (IRB # STU24696) and performed in accordance with the Declaration of Helsinki. Freshly collected tumor specimens were

dissected for single cell dissociation, formalin fixation with subsequent paraffin embedding, or flash frozen.

Blood was collected via phlebotomy following informed consent as above and peripheral blood mononuclear cells (PBMCs) isolated under Ficoll gradient and cryopreserved in ~1–2e6 cell aliquots.

scRNA-seq tissue sample processing:

Tissue dissociation: Human skin tumor biopsies or portions of surgical resections were dissociated into single-cell suspensions using either the Miltenyi Biotec human tumor dissociation kit (enzymes H, R, A) or a mixture of RPMI, DNase (0.1 mg/mL), collagenase IV (2mg/mL) and hyaluronidase (0.1mg/mL). Biopsies were gently minced into 1–2 mm pieces, transferred to an enzyme mix as above, and incubated at 37°C for 30–60 min at 800 rpm in a thermomixer or by gentle rocking. Following incubation, cells were filtered, washed, and cryopreserved.

Cryopreserved, digested MCC tumor samples were thawed and processed following a modified version of the 10x Demonstrated Protocol “Thawing Dissociated Tumor Cells for Single Cell RNA Sequencing” and BioLegend TotalSeq C protocol. The samples were thawed, labeled with TotalSeq C antibodies, and stained with 7-AAD viability staining solution and Apotracker Green (BioLegend). Cells were then subjected to FACS to sort the CD45+ and CD45- cell populations for target enrichment of 2/3 CD45+ cells. After sorting, the final cell preparation was resuspended in RPMI + 10% FBS + SUPERase•In™ RNase Inhibitor (Sigma), filtered, and counted to ensure an optimal concentration of 1e6 cells/mL. Finally, the cell suspension was processed for encapsulation and reverse transcription using the 10× 5' v2 Chromium platform, according to the manufacturer's instructions.

scRNA-seq PBMC sample processing:

Cryopreserved peripheral blood mononuclear cells (PBMCs) were thawed, labeled, and prepared for 10x droplet encapsulation following a modified protocol adapted from the 10x and BioLegend TotalSeq C protocols. Cells were blocked using Human TruStain FcX™ Fc blocking reagent (BioLegend) and stained with TotalSeq™ C and hashing antibodies (BioLegend). Live cells were sorted using fluorescence-activated cell sorting (FACS) with a gate set for CD45+ and CD3+ cells. The sorted cells were combined with CD3- cells to maintain a 1:3 ratio and cryopreserved in Bamberger preservation medium with SUPERase•In™ RNase Inhibitor (Sigma). Equal volumes of four to five hashed samples were combined to ensure that samples stained with the same hashing antibody were not mixed. The cells were slow-frozen at –80°C before being transferred to liquid nitrogen for long-term storage. Thawing and final sample preparation for encapsulation were performed immediately before loading onto the 10× 5' v2 Chromium platform. Sequencing parameters were set for the 5' cDNA, αβ VDJ, γδ VDJ, and ADT libraries according to the targeted cell number for single-cell RNA sequencing.

T cell receptor sequencing:

$\alpha\beta/\gamma\delta$ TCR libraries were prepared following the 10x Genomics Single Cell V(D)J protocol. TCR $\gamma\delta$ libraries were prepared following the 10x Genomics protocol to generate TCR $\alpha\beta$ libraries with modifications. Specific primers were designed, including R1_hTRDC (5'AGCTTGACAGCATTGTACTTCC), R1_hTRGC (5'TGTGTCGTTAGTCTTCATGGTGTTC), R2_hTRDC (5'TCCTTCACCAGACAAGCGAC), R2_hTRGC (5'GATCCCAGAATCGTGTGCTC), and the forward primer (5'GATCTACACTCTTCCCTACACGACGC), which were resuspended in 100 μ M EB buffer. The library preparation involved two PCR reactions: PCR1, using Human $\gamma\delta$ mix 1 (R1_hTRDC + R1_hTRGC), and PCR2, using Human $\gamma\delta$ mix 2 (R2_hTRDC + R2_hTRGC), substituting the primers as detailed. PCR1 was performed using 2–5 μ L full-length cDNA, 50 μ L 2x Kapa HIFI Master Mix, 5 μ L cDNA additive, 2 μ L 20 μ M forward primer, and 2 μ L 20 μ M human $\gamma\delta$ mix 1, with a reaction volume of 100 μ L using water. The PCR was performed for 14 cycles. PCR2 was performed with 35 μ L bead elution, 50 μ L 2x Kapa HIFI Master Mix, 5 μ L cDNA additive, 2 μ L 20 μ M forward primer, and 2 μ L 20 μ M human $\gamma\delta$ mix 2, with a reaction volume of 100 μ L using water. The PCR was performed for 12 cycles. Cleanups and PCR conditions were identical to the 10x protocol, but $\gamma\delta$ libraries required additional amplification cycles because of the rarity of $\gamma\delta$ T cells compared to $\alpha\beta$ T cells in most cell populations. Finally, TCR $\gamma\delta$ enriched libraries were processed according to the 10x Genomics Single Cell V(D)J protocol.

FFPE RNA-seq:

Formalin-fixed paraffin-embedded (FFPE) tissue blocks were sectioned at a thickness of 5 μ m. RNA was extracted from FFPE tissue sections using the RNeasy FFPE Kit (Qiagen) according to the manufacturer's instructions. The extracted RNA was treated with an RNase-Free DNase Set (Qiagen) to remove any genomic DNA contamination. RNA quality and quantity were assessed using a Bioanalyzer (Agilent) and Qubit fluorometer (Thermo Fisher Scientific), respectively.

RNA sequencing libraries were prepared using the TruSeq Stranded mRNA LT Sample Prep Kit (Illumina), according to the manufacturer's instructions. Briefly, 500 ng total RNA was used for poly(A) mRNA selection, fragmentation, and cDNA synthesis. The resulting cDNA was end-repaired, adenylated, and ligated using Illumina-specific adapters. PCR amplification was performed to enrich adapter-ligated fragments. The final library was purified and validated for size distribution using a Bioanalyzer (Agilent) and quantified by qPCR using a Kapa Library Quantification Kit (Kapa Biosystems). Libraries were pooled at equimolar concentrations and sequenced using an Illumina NovaSeq 6000.

Fluorescent Multiplex Immunofluorescence Panel Procedure:

Formalin-fixed and paraffin-embedded (FFPE) tissue samples were immunostained using the AKOYA Biosciences OPAL TM 7-Color Automation IHC kit (Waltham, MA, USA) on a BOND RX autostainer (Leica Biosystems, Wetzlar, Germany). The OPAL 7-color kit uses tyramide signal amplification (TSA)-conjugated to individual fluorophores to detect various targets within the multiplex assay.

The sections were baked at 65°C for one hour then transferred to BOND RX (Leica Biosystems). All subsequent steps (deparaffinization and antigen retrieval) were performed using an automated OPAL IHC procedure (AKOYA). OPAL staining of each antigen was performed as follows: heat-induced epitope retrieval (HIER) was achieved with EDTA buffer (pH 9.0) for 20 min at 95°C before the slides were blocked with AKOYA blocking buffer for 10 min. The slides were then incubated with the primary antibody, GZMB (CST, D6E9W, RRID:AB_2799313, 1:75, dye 570) at RT for 30 min, followed by incubation with OPAL HRP polymer and one of the OPAL fluorophores during the final TSA step. Individual antibody complexes were stripped after each round of antigen detection in an optimized order and were determined independently for each panel. After the final stripping step, DAPI counterstain was applied to the multiplexed slide and removed from the BOND RX for coverslipping with ProLong Diamond Antifade Mountant (Thermo Fisher Scientific). All slides were imaged using the Vectra3 Automated Quantitative Pathology Imaging System.

Quantitative Image Analysis:

Multilayer TIFF images were exported from InForm (AKOYA) and loaded into HALO (Indica Labs, New Mexico) for quantitative image analysis. A classifier was trained to identify areas of the tumor, stroma, or non-tissue regions. Pan-cytokeratin was used to train tumor regions because it is a masking marker for tumor cells. A classifier was created and tested on various images of the image set. The tissue was segmented into individual cells using the DAPI marker, which stains the cell nuclei. For each marker, a positivity threshold within the nucleus or cytoplasm was determined per marker based on published staining patterns and intensities for that specific antibody. After setting a positive fluorescent threshold for each staining marker, the entire image set was analyzed using the created algorithm. The generated data included positive cell counts for each fluorescent marker in the cytoplasm or nucleus and the percentage of cells positive for the marker. The data was sorted by classification to include the entire tissue, stroma, or tumor regions. Along with the summary output, per-cell analysis was exported to provide the marker status, classification, and fluorescence intensities of every individual cell within an image.

Local density analysis was performed by outlining the bulk tumor based on para-nuclear dot-like accentuation of panCK expression. Epidermal/follicular expression of panCK was excluded from analysis. Following the definition of the tumor-stromal interface, bands of 40 µm widths were defined as +500 microns (out of tumor into stroma) and -500 microns (into tumor interior), and the local density of cell counts according to marker expression was calculated per mm².

Spatial Transcriptomics Nanostring GeoMx:

Tissue slices of 15 MCC tumor samples from 12 patients were placed on charged glass slides and stained with four immunofluorescent markers to assist in the selection of 12 regions of interest (ROIs) per sample. The tissue was selected from a cohort of patients with MCC who received immunotherapy (four responders and eight non-responders). FFPE human skin samples were baked for 1 h at 65°C and processed on the Leica automation platform in three major steps: 1) slide baking and deparaffinization, 2) antigen retrieval for 20 min at 100°C, 3) 1.0 µg/ml Proteinase K treatment for 15 min. Slides were incubated

overnight with GeoMx CTA or WTA assay probe cocktail. The following day, the slides were washed and subjected to morphological marker incubation before loading onto a GeoMx machine for fluorescence scanning and ROI selection. Barcodes were collected, and the subsequent barcoding reading was completed on the Illumina NGS platform.

The markers used in the ROI selection were SYTO13 (DNA/nuclei), KRT18 (keratin/cancer cells), PTPRC (CD45/lymphocytes), and CD3E (T cells).

Reagent: Syto13

Manufacturer: Thermo Fisher

ID: 121301310

Protein: PanCK

Manufacturer: Novus

Clone: AE1/AE3

ID: NBP2-29429, RRID:AB_3068002

Protein: CD45

Manufacturer: CST

Clone: D9M8I

ID: 13917BF, RRID:AB_2750898

Protein: CD3

Manufacturer: Origene

Clone: UMAB54

ID: UM500048, RRID:AB_2629062

The classifications assigned to the regions of interest (ROIs) were based on manual examination by the pathologist in our team, aided by DAPI and PanCK immunofluorescent stain, as presented in Supplementary Fig. S18A. Selection of the ROIs was done by the pathologist before RNA quantitation by GeoMx. Those ROIs in which the pathologist observed a predominance of epithelial cells were classified as tumor ROIs. Contrarily, those ROIs showing the absence of epithelial cells were classified as stroma. Variations of these ROIs were also included, in which tumor or stroma ROIs could also present infiltration of immune cells (CD45 or CD3 stain). ROIs of arbitrary shapes were selected to represent the tumor and stroma, including areas close to the tumor-stromal interface and central tumor areas corresponding to at least 150 cells. assigned to five categories with respect to their spatial location and level of immune infiltration:1) “immune cold” stroma (paucity of immune cells), 2) immune-excluded stroma (immune cells distant from tumor

nodule), 3) immune-infiltrated stroma (stroma adjacent to tumor infiltrated by immune cells), 4) tumor close to immune cells (adjacent tumor areas to immune-infiltrated stroma), and 5) inner tumor (central areas of tumor nodules typically devoid of significant immune infiltrates) (Supplementary Fig. S17A). The slides were profiled for gene expression using the Nanostring Cancer Transcriptome Atlas panel (CTA, 9 samples/108 ROIs) and Whole Transcriptome Atlas panel (WTA, 6 samples/72 ROIs). The number of reads per ROI varied from 1.1×10^6 to 142.9×10^6 (Supplementary Fig. S17B)

The resulting .dcc files were processed using the GeoMx Analysis Suite v2.5. Probe counts were filtered using the default values and aggregated at the gene level. Given the technical effects associated with the panel used (WTA or CTA), we applied ComBat correction as implemented in the R package sva. We applied 3rd-quartile normalization of the counts, followed by log transformation. Normalized counts were used to detect differentially expressed genes between immune infiltration categories and patient therapy responses within each spatial niche (stroma/tumor). Differential gene expression was performed via pairwise t-tests in the scan package, and p-values were adjusted for multiple testing using false discovery rate (FDR).

Spatial Transcriptomics CosMx Spatial Molecular Imaging (SMI):

To study the spatial patterns in gene expression of immune and tumor cells, we profiled six slides containing MCC tissue using the 960-gene Immune Oncology panel Nanostring's SMI platform. Tissue samples were obtained from 12 patients, six of which showed a positive response to therapy, four were non-responders and two were not accessible. Formalin-fixed, paraffin-embedded (FFPE) tissue sections were prepared for CosMx SMI profiling as described by He *et al* (45). Morphological markers used.

Protein: CD298

Manufacturer: Abcam

ID: ab167390, RRID:AB_2890241

Protein: B2M

Manufacturer: Abcam

ID: ab214769, RRID:AB_1266996

Protein: PanCK

Manufacturer: Novus

ID: NB2-33200AF532, RRID:AB_2924722

Protein: CD45

Manufacturer: Novus

ID: NBP2-34528AF594, RRID:AB_960384

Protein: CD3

Manufacturer: Origene

ID: UM500048, RRID:AB_2629062

A total of 122 fields of view (FOVs) were profiled (2–16 per patient), which were chosen by inspecting adjacent H&E-stained tissue slices (Supplementary Fig. S19A) and reconfirmed by imaging with morphological markers. The raw dataset consisted of 712,617 image-segmented cells, and transcripts falling within the cells were summarized at the gene level. Cells without any recorded transcripts were removed from the downstream analyses (815 cells). The SMI platform includes 19 negative probes for assessment to assist in the removal of background levels for transcript detection and quality control. We removed from the data set cells with more than 5% negative probe transcripts as well as cells with less than 50 expressed genes. To reduce the effect of segmentation artifacts (i.e., two or more cells merged into a single segment – “doublets”), we removed cells with more than 500 expressed genes or 1250 transcript counts. The filtering procedure resulted in 614,209 cells being used in downstream analyses. We applied SCTransform normalization to the resulting counts. Corrected counts were input to InSituTypeML, a Bayes classifier to transfer the cell types identified in the tissue single-cell dataset to the SMI cells. To account for the background signal, InSituTypeML used the average negative probe count per cell. We also inferred Louvain clusters and tested for differentially expressed genes among clusters using Seurat to manually curate annotations generated by InSituTypeML. We confirmed that DE genes from the assigned CosMx clusters significantly overlapped with the DE genes from the higher resolution single cell dataset (Supplementary Fig. S19D).

To quantify the degree of dispersion or co-localization of specific cell types within the spatial immune landscape in the MCC samples, we used spatialTIME(34) to measure the degree of co-localization (DoC) across all possible binary combinations of cell subsets based on scRNA-seq data. The method implemented in spatialTIME computes the bivariate Ripley’s K statistic and assumes that cells occur at the same frequency across an FOV (i.e., complete spatial randomness), and is further normalized for area and cell numbers. Because FOVs often show non-uniform cellularity, spatialTIME compares the observed K to the exact K estimate based on a spatial random point process. This difference is the derived measure, called the degree of co-localization.

MCPyV epitope screen:

Autologous lymphoblastoid cell lines (LCLs) were generated using a modified version of the *Current Immunology* protocol 7.22. Peripheral blood mononuclear cells (PBMCs) were thawed and washed with prewarmed complete RPMI medium. After counting and assessing viability, cells were centrifuged and resuspended in complete RPMI with 1.1 µg/mL cyclosporin A concentration of 2.2×10^6 cells/mL. The cell suspension was incubated for 1 h at 37°C before adding the thawed Epstein-Barr virus (EBV) viral supernatant (ATCC) to achieve a final cell concentration of 2×10^6 cells/mL, with the virus constituting 10% of the final volume. The culture was monitored daily, and when the medium turned yellow, it was gently aspirated and replaced with fresh RPMI medium containing 1 µg/mL cyclosporin

A. After 2–3 weeks, large cell clumps appeared and cyclosporin was no longer added to the culture. LCLs were maintained at high densities ($>1 \times 10^6$ cells/mL) and thrived in slightly acidic (yellowish) media, with pipetting used to break up large cell clumps, as needed. After expansion, the LCLs were cryopreserved for further use.

The patient's LCLs were thawed and treated with Mitomycin C to inhibit proliferation, followed by washing and plating. Jurkat-76 cells previously transduced with a CD8-NFAT-GFP reporter construct (RRID:Addgene_153417)(46) were lentivirally transduced with transgenic TCRs (TWIST Biosciences) and selected in 1 μ g/mL puromycin. 116, 13-mer peptides were designed to tile along the small T and large T antigens of MCPyV with 8-mer overlap (Genscript)(24) (Supplementary Table S23). Peptides were pooled into 22 pools, such that each peptide was present in two pools. On day 2, the peptide pools were added to the LCLs and incubated for an hour at 37°C. Peptide-loaded cells were washed with cRPMI. Jurkat reporters were added to each LCL well, mixed, and centrifuged before incubation for 20–24 hours at 37°C. The next day, the remaining cells were stained with CD69 PE (BioLegend Cat# 310905 (also 310906), RRID:AB_314840), HLA-DR APC-Cy7 (BioLegend Cat# 307618 (also 307617), RRID:AB_493586), CD3 BV421 (BioLegend Cat# 344834 (also 344833), RRID:AB_2565675) antibodies for 30 min. The cells were washed, resuspended with 7AAD, and analyzed using a Fortessa HTS flow cytometer to measure Live, HLA-DR-, CD3+ CD69+ GFP+ cells. Jurkat-76 cells were received as a gift from Mirjam H. M. Heemskerk on 09/25/2018(47). They were authenticated as TCR-negative, but no additional authentication was performed. Upon initial culture, the cells were periodically tested for mycoplasma with the MycoAlert kit (Lonza). Cultures were passaged up to 5 times before experiments.

Computational methods:

scRNA-seq pre-processing: The GRCh38.p13 primary assembly from Ensembl Release 98 and the GENCODE v32 primary assembly were modified as follows. Cellranger 6.1.0 was used to create a reference for scRNA-seq alignment. The MCPyV sequence was then added to the primary and final GTF files.

FASTQ files containing gene expression and feature barcodes (ADT) were aligned to the human hg38 and MCPyV genomes using CellRanger 6.1.0. Ambient RNA was removed using the CellBender pipeline(48). Processed read counts for each sample were imported as Seurat objects, after which TCR-seq data were added. We then filtered the data based on mitochondrial reads (25% cut-off) and minimum RNA genes (200 genes). Doublets were removed using DoubletFinder with the default pN of 0.25 and the optimal pK value calculated with the “find.pK” function. Cells were annotated with ProjectTILS and SingleR using Monaco and HPCA reference datasets. Data were merged into a single Seurat object. We performed standard Seurat (4.0) clustering, Harmony batch correction, and manual cell cluster annotation based on marker genes, including transcription factors, effector molecules, and previously annotated cell types. The data were sub-clustered by major cell types. Contaminating cells were filtered based on marker gene expression. Differential gene expression, gene enrichment, and GSVA were performed using the default Seurat pipeline,

followed by comparison of cell type proportions, analysis of TCR clonotype expression, and TCR linkage analysis.

Enrichr gene ontology analysis: Protein-coding genes were subset upregulated or downregulated genes as indicated and input into the Enrichr pipeline using the enrichR package (3.0) and querying all available databases(9). The resulting enriched gene sets were grouped by database and then sorted by “combined score” which was labeled the Enrichr score.

Pseudobulk differential gene expression analysis: The cells of interest were subset from the full single-cell RNA-seq object. Cells were then grouped by sample into a pseudobulk matrix using Seurat’s “AggregateExpression” function. Next, this matrix was used input into differential expression analysis using edgeR (3.36.0) using the “glmLRT” function.

Gene set enrichment analysis (GSEA): Genes from differential expression analysis were ranked by signed p-value. These genes were used as input for the fgsea function from the fgsea package (1.24). Reference genesets were derived from all human geneset in the msigdb package (7.5.1). To control for cell cycle and tumor cycling cells, GSEA was run on two different pseudobulk DE results on tumor cells. The first result was without any tumor cycling cells and the second was controlling for cell cycling scoring as described in the pseudotime analysis and covariate DE-seq.

Statistical Considerations for RNA-seq analysis

In the absence of specific hypothesis testing, sample size rationales were not calculated a priori. The major components of data were bulk RNA-seq and scRNA-seq data. These samples numbered 85 (63 for pre-ICB analysis) and 109, respectively. For bulk RNA-seq, primary conclusions were based upon signature based analyses and the power calculation for the RNAseq experiment was conducted using package ‘ssizeRNA’. With an assumption of 10,000 tested genes (including the top 100 prognostic genes), a minimum fold change criterion of 2, and accounting for dispersion parameters ranging from 0.1 to 0.4, 63 samples (comprising 32 responders and 31 non-responders) achieves an estimated probability of more than 80% (desired power) to reject the null hypothesis assuming equality in expression means between the two groups, with an associated false discovery rate (FDR) of 0.05. Additionally, we performed power calculations based on the gene set-based signature, specifically GSVA scores. Assuming a normal distribution of GSVA scores, our sample size of 63 provides a power of 90% to detect effect sizes (Cohen’s d) at 0.86 and a power >80% to detect effect sizes at 0.75, while assuming a type I error rate of 0.05. In our single-cell RNA-seq experiments, where each sample comprises an average of 5,129 cells. Assuming that each subpopulation contains a minimum of 10 cells, our approach achieves a 95% probability of success to detect cell subpopulations with frequency as rare as 0.003.

Gene set module score: Cells in the single-cell RNA-seq dataset were assigned a module score using the “AddModuleScore” function in the Seurat package. This function was run on the combined Seurat object with all samples. In some cases, the cells were

grouped by sample and/or cell type, and the average module score was reported. The following gene sets were used: Hallmark interferon gamma signature (MSIGDB), Hallmark interferon alpha signature (MSIGDB), exhaustion score from the cited paper(37), and NeoTCR8 score from the cited paper(23).

SCENIC analysis: Following loom file conversion from the subset Seurat object, we executed the standard pySCENIC pipeline (v 0.12.1)(15) in three major steps: gene regulatory network inference, context-dependent enrichments using cis-regulatory motifs, and activity scoring of regulons in individual cells using the AUCell algorithm. Subsequent to the SCENIC analysis, we integrated the results into our Seurat object. We used the FindAllMarkers function, ensuring even sample representation with equal number of cells, to find differentially expressed regulons between responders and non-responders.

Diffusion mapping: Cells were re-clustered with cell cycle genes regressed using Seurat's "CellCycleScoring" and "ScaleData" functions. Batch-corrected principal components were input into the DiffusionMap function of the Destiny package (3.14).

Pseudotime analysis: Cells were subset and re-clustered with cell cycle genes regressed using Seurat's "CellCycleScoring" and "ScaleData" functions. After cell cycle regression, pseudotime was determined using the default pipeline in monocle3 (1.0.0). The "root" node for pseudotime was determined by expression of naïve like markers including CD45RA, *TCF7*, and non-expression of exhausted or memory markers including PD-1, CD45RO, *TOX*. Genes were then linearly correlated with pseudotime using Pearson correlation. The next genes were non-linearly correlated to pseudotime using the "fitGAM" function in the tradeSeq (1.8.0) package. The expression levels of significant genes were scaled across pseudotime and smoothed using the "smooth.spline" function with five degrees of freedom from the stats package (4.1.1). Genes were ordered by maximum value across pseudotime and then smoothed/scaled expression was plotted as a heatmap using the ComplexHeatmap package (2.13.2, RRID:SCR_017270).

CellChat: All cells from the tissue scRNA-seq dataset were made into a pseudobulk average expression Seurat object, grouped by sample and cell type. This was inputted into the CellChat (1.4.0) pipeline using the human database accessed on April 29th, 2023.

Projection of additional post-ICB scRNA-seq samples: Five new post-ICB samples were harmonized using the Harmony package. All cells were mapped to the original UMAP and clusters from the original 49 sample tissue MCC scRNA-seq data using the Seurat "FindTransferAnchors" and "MapQuery" functions.

Single cell TCR-seq pre-processing: FASTQ files for both $\alpha\beta/\gamma\delta$ VDJ sequencing were input into the Cellranger (3.0.2) vdj pipeline. Both contig annotations and clonotype annotations were added as metadata to the scRNA-seq Seurat objects for downstream analysis.

TCR-linkage analysis: Identical clonotypes were matched between cell clusters and the likelihood ratio of clonotype matching was calculated for each cluster pair. P-values

were determined using Fisher's test and adjusted using the Benjamini-Hochberg Procedure. Finally, log-likelihood ratios were subjected to unsupervised clustering and plotted using the Pheatmap (v1.0.12) package.

Monte-Carlo simulation of clonotype restriction: All CD8 T cells were grouped according to a unique TCR clonotype. Each cell was randomly assigned either Tcirc or Trm. The number of clonotypes that were 100% restricted to either Trm or Tcirc was counted. This simulation was repeated 1e6 times. The p-value was determined by the proportion of simulations that had equal or greater numbers of restricted clonotypes than what was observed in the dataset.

GLIPH2 analysis: TCR clonotype information was extracted from the CD8 T cells. Additionally, samples were HLA-typed using Polysolver, Optitype, ArcasHLA, and HLA-HD from bulk and single RNA-seq data. These HLA-types and the extracted clonotype information were input into the GLIPH2 online portal (<http://50.255.35.37:8080>) with the following parameters: GLIPH2 algorithm, reference version 2.0, CD8 reference, all_aa_interchangeable as "YES." The results were filtered using the Fisher_score <0.05, number_unique_CDR3 > number_subject, and Freq > number_unique_CDR3.

CD8 clonotype database comparison: TCR clonotype information was extracted from CD8 positive T cells: the TRB CDR3 and *TRBV* genes. TCRs were considered a match if both the TRB CDR3 and *TRBV* genes matched a TCR found in the database. The VDJdb, McPAS-TCR, and TRAdb databases were accessed on January 2nd, 2023(27)(28)(29).

γδ TCR clustering: The delta CDR3 sequences were extracted from the scTCR dataset. All CDR3s were aligned pairwise to each other using the "pairwiseAlignment" function from the Biostrings package (2.62.0) with the following settings: BLOSUM62 substitution matrix, gap opening = -10, gap extension = -1. The pairwise alignments were then clustered using unsupervised hierarchical clustering.

MIXCR: We implemented the MIXCR software(36) (4.1.0) to extract T/B cell receptor sequences from the RNA-seq data. The command used was mixcr to analyze RNA-seq- tcr-full-length, with the parameters --species hsa to specify the human species and --rna to indicate RNA input data. We included the --keep-non-CDR3-alignments flag to retain alignments that did not include the CDR3 region, which provides a more comprehensive perspective on the TCR repertoire. The input data consisted of paired-end RNA-seq reads. Expanded clones were defined as clones with greater than 1.5 times the reads of the least frequent clone per sample and per receptor type.

Bulk RNA-seq processing: Sequence reads were trimmed and filtered using Trim Galore! (v0.6.7, RRID:SCR_011847), to remove low-quality reads, adapter sequences, and bases with low Phred scores. The reads were then aligned to the reference human genome (GRCh38.p13 primary assembly) augmented with the Merkel cell polyomavirus transcriptome using GENCODE v32 transcript annotation with STAR (v2.7.10b, RRID:SCR_004463) in two-pass mode. The dataset was quality controlled to remove any significant outliers. Differential gene expression analysis was performed using

DESeq2 (1.28.1) (RRID:SCR_000154) or other appropriate software packages. The limma package (RRID:SCR_010943) was used to fit the differential expressions to a generalized linear model. Pathway analysis was conducted with GSEA and gene set enrichment analysis (GSEA).

Covariate DESeq: The patient metadata associated with the corresponding RNA-seq were associated with a DESeq object (RRID:SCR_000154). Raw read counts were filtered to remove low-expression genes, defined as having more than 20% of the total samples containing zero reads. A variance-stabilizing transformation was applied to the DESeq object and a PCA plot was visualized, showing clustering by virus status, tissue of origin, and patient sex. Differential expression was determined by comparing the immunotherapy response with the covariates virus status, tissue of origin, and patient sex. For pre-post ICB DE-seq, only patients were used as covariates with samples compared by time point.

Cell-type signature generation: The Wilcoxon markers of the cell type of interest versus all other cells were compared to the Wilcoxon markers of that cell type compared to the next most similar cell type. Trm cells were compared to Tcirc cells and tumor cycling cells were compared to all other tumor cells. These markers were intersected and filtered for a log2 fold change > 0.5 and a p value < 0.05. The genes are listed in Supplementary Table S17.

Gene set score: The cell-type signature genes were used as a gene set for single-sample GSEA. The corresponding GSEA scores were then classified into quartiles. The Kaplan-Meier curve for immunotherapy-specific survival compared to the top 25% of the cell-type score against the bottom 25% cell-type score among patients who had an evaluable immunotherapy response.

TCR-pMHC complex prediction: TCR-pMHC complex models were generated using TCRmodel2(25) and AlphaFold(48) (v.2.3.0) deep learning structural prediction methods. Structural prediction using TCRmodel2 was performed via its online server (<https://tcmodel.ibbr.umd.edu/>), with structural refinement. AlphaFold 2.3.0 was downloaded from the AlphaFold Github repository (<https://github.com/deepmind/alphafold>) in February 2023 and run on a local computing cluster, generating 25 predictions per complex, with refinement performed on the top-ranked predictions. No template date cutoff was applied during TCRmodel2 or AlphaFold modeling. For the MO_MCC_095c1 TCR-peptide-MHC complex (Fig. 4D), models from TCRmodel2 and AlphaFold were pooled and ranked using the AlphaFold model confidence score, which is associated with TCR-pMHC model accuracy(25). The AlphaFold-generated model was ranked the highest and was selected for further inspection. For all other TCR-peptide-MHC complexes, models were generated by TCRmodel2 and ranked by model confidence score.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Acknowledgements

This research was made possible through the Division of Research Pathology, Total Cancer Care protocol and supported by the Tissue Core at the H. Lee Moffitt Cancer Center & Research Institute, an NCI-designated Comprehensive Cancer Center (P30-CA076292). Funding support was provided by the Melanoma & Skin Cancer Center of Excellence (K.Y.T.), the Barry S. Greene Fund (K.Y.T, A.B.), and the Moffitt Distinguished Scholar Award (N.I.K.). The authors acknowledge the Nanostring Technology Access Program for access to both the GeoMx and CosMx platforms to generate spatial transcriptomic data. We would like to thank the Northwestern Dermatology Clinical Trials Unit for their assistance with tissue acquisition. This work was supported by the Northwestern University Flow Cytometry Core Facility and the Cancer Center Support Grant (NCI CA060553). The research reported in this publication was supported by the Northwestern University Skin Biology & Diseases Resource-Based Center of the National Institutes of Health under the award number P30AR075049. Funding for this study was provided by the National Cancer Institute grant T32 CA009560 and F30CA278298 (Z.Z.R.), the 2019 AOA Carolyn E Kuckein Fellowship (Y.Z.), the V Foundation for Cancer Research grant T2021-019 (J.C.), the Bakewell Foundation (J.C.), National Institutes of Health grants DP2 OD024475-01 (J.C.), R35 GM144083 (B.G.P.), and the Leukemia and Lymphoma Society grant 1377-21 (J.C.).

Data Availability

Bulk RNA-seq and scRNA-seq data is available at GEO accession GSE235093.
Raw sequencing data is available at dbGAP accession phs003629 (<https://www.ncbi.nlm.nih.gov/gap/sstr/report/phs003629.v1.p1>).

Abbreviations list:

MCC	Merkel cell carcinoma
ICB	immune checkpoint blockade
VP	virus positive
VN	virus negative
MCPyV	Merkel cell polyomavirus
mIF	multiplexed immunofluorescence
FFPE	Formalin-fixed paraffin-embedded
LCLs	lymphoblastoid cell lines
EBV	Epstein-Barr virus
PBMCs	peripheral blood mononuclear cells
FACS	fluorescence-activated cell sorting
ADT	antibody-derived tags
DE	differentially expressed
FDR	false discovery rate
MSIGDB	Molecular Signatures Database
KEGG	Kyoto Encyclopedia of Genes and Genomes

ENCODE	Encyclopedia of DNA Elements
ChEA	ChIP Enrichment Analysis
UMAP	Uniform Manifold Approximation and Projection
GSEA	Gene set enrichment analysis
NES	normalized enrichment score
TME	tumor microenvironment
TSA	tyramide signal amplification
LLR	log-likelihood ratio
TCR	T cell receptor
CDR3	complementarity-determining regions 3
ROIs	Regions of Interest
CTA	Cancer Transcriptome Atlas
WTA	Whole Transcriptome Atlas
FOVs	fields of view
DoC	degree of clustering
Tcirc	circulating T cells
Trm	tissue-resident memory T cells
Temra	terminally differentiated effector memory
GSVA	gene set variance analysis
CR	complete response
PR	partial response
SD	stable disease
PD	progressive disease
NK	natural killer
pDC	plasmacytoid dendritic cell
cDC	classical dendritic cell
KLR	killer cell lectin-like receptor
KIR	killer cell immunoglobulin-like receptor
IGH	immunoglobulin heavy chain

TRB	TCR beta chain
LN	lymph node
MHC	major histocompatibility complex

References

- McEvoy AM, Lachance K, Hippe DS, Cahill K, Moshiri Y, Lewis CW, et al. Recurrence and Mortality Risk of Merkel Cell Carcinoma by Cancer Stage and Time From Diagnosis. *JAMA Dermatol.* 2022;158:382–9. [PubMed: 35195657]
- Knepper TC, Montesion M, Russell JS, Sokol ES, Frampton GM, Miller VA, et al. The Genomic Landscape of Merkel Cell Carcinoma and Clinicogenomic Biomarkers of Response to Immune Checkpoint Inhibitor Therapy. *Clin Cancer Res.* 2019;25:5961–71. [PubMed: 31399473]
- DeCaprio JA. Molecular Pathogenesis of Merkel Cell Carcinoma. *Annu Rev Pathology Mech Dis.* 2020;16:1–23.
- Goh G, Walradt T, Markarov V, Blom A, Riaz N, Doumani R, et al. Mutational landscape of MCPyV-positive and MCPyV-negative Merkel cell carcinomas with implications for immunotherapy. *Oncotarget.* 2016;7:3393–405.
- Lahman MC, Paulson KG, Nghiem PT, Chapuis AG. Quality Is King: Fundamental Insights into Tumor Antigenicity from Virus-Associated Merkel Cell Carcinoma. *J Invest Dermatol.* 2021;141:1897–905. [PubMed: 33863500]
- Nghiem P, Bhatia S, Lipson EJ, Sharfman WH, Kudchadkar RR, Brohl AS, et al. Three-year survival, correlates and salvage therapies in patients receiving first-line pembrolizumab for advanced Merkel cell carcinoma. *J Immunother Cancer.* 2021;9:e002478.
- D'Angelo SP, Lebbé C, Mortier L, Brohl AS, Fazio N, Grob J-J, et al. First-line avelumab in a cohort of 116 patients with metastatic Merkel cell carcinoma (JAVELIN Merkel 200): primary and biomarker analyses of a phase II study. *J Immunother Cancer.* 2021;9:e002466.
- Paulson KG, Voillet V, McAfee MS, Hunter DS, Wagener FD, Perdicchio M, et al. Acquired cancer resistance to combination immunotherapy from transcriptional loss of class I HLA. *Nat Commun.* 2018;1–10. [PubMed: 29317637]
- Chen EY, Tan CM, Kou Y, Duan Q, Wang Z, Meirelles GV, et al. Enrichr: interactive and collaborative HTML5 gene list enrichment analysis tool. *Bmc Bioinformatics.* 2013;14:128. [PubMed: 23586463]
- Grasso CS, Tsoi J, Onyshchenko M, Abril-Rodriguez G, Ross-Macdonald P, Wind-Rotolo M, et al. Conserved Interferon- γ Signaling Drives Clinical Response to Immune Checkpoint Blockade Therapy in Melanoma. *Cancer Cell.* 2020;38:500–515.e3. [PubMed: 32916126]
- Huang R, Grishagin I, Wang Y, Zhao T, Greene J, Obenauer JC, et al. The NCATS BioPlanet – An Integrated Platform for Exploring the Universe of Cellular Signaling Pathways for Toxicology, Systems Biology, and Chemical Genomics. *Front Pharmacol.* 2019;10:445. [PubMed: 31133849]
- Tirosch I, Izar B, Prakadan SM, II MHW, Treacy D, Trombetta JJ, et al. Dissecting the multicellular ecosystem of metastatic melanoma by single-cell RNA-seq. *Science.* 2016;352:189–96. [PubMed: 27124452]
- Reizis B. Plasmacytoid Dendritic Cells: Development, Regulation, and Function. *Immunity.* 2019;50:37–50. [PubMed: 30650380]
- Hegde PS, Chen DS. Top 10 Challenges in Cancer Immunotherapy. *Immunity.* 2020;52:17–35. [PubMed: 31940268]
- de Sande BV, Flerin C, Davie K, Waegeneer MD, Hulselmans G, Aibar S, et al. A scalable SCENIC workflow for single-cell gene regulatory network analysis. *Nat Protoc.* 2020;15:2247–76. [PubMed: 32561888]
- Coughlan E, Garside VC, Wong SFL, Liang H, Kraus D, Karmakar K, et al. A Hox Code Defines Spinocerebellar Neuron Subtype Regionalization. *Cell Rep.* 2019;29:2408–2421.e4. [PubMed: 31747609]

17. Nefzger CM, Haynes JM, Pouton CW. Directed Expression of Gata2, Mash1, and Foxa2 Synergize to Induce the Serotonergic Neuron Phenotype During In Vitro Differentiation of Embryonic Stem Cells. *STEM CELLS*. 2011;29:928–39. [PubMed: 21472823]
18. Bunt J, Hasselt NE, Zwijnenburg DA, Hamdi M, Koster J, Versteeg R, et al. OTX2 directly activates cell cycle genes and inhibits differentiation in medulloblastoma cells. *Int J Cancer*. 2012;131:E21–32. [PubMed: 21964830]
19. van der Leun AM, Thommen DS, Schumacher TN. CD8+ T cell states in human cancer: insights from single-cell analysis. *Nat Rev Cancer*. 2020;20:218–32. [PubMed: 32024970]
20. Yenyuwadee S, Lopez JLS-T, Shah R, Rosato PC, Boussiotis VA. The evolving role of tissue-resident memory T cells in infections and cancer. *Sci Adv*. 2022;8:eabo5871.
21. Tokura Y, Phadungsaksawasdi P, Kurihara K, Fujiyama T, Honda T. Pathophysiology of Skin Resident Memory T Cells. *Front Immunol*. 2021;11:618897.
22. Heczey A, Xu X, Courtney AN, Tian G, Barragan GA, Guo L, et al. Anti-GD2 CAR-NKT cells in relapsed or refractory neuroblastoma: updated phase 1 trial interim results. *Nat Med*. 2023;1–10. [PubMed: 36694061]
23. Lowery FJ, Krishna S, Yossef R, Parikh NB, Chatani PD, Zacharakis N, et al. Molecular signatures of antitumor neoantigen-reactive T cells from metastatic human cancers. *Science*. 2022;375:877–84. [PubMed: 35113651]
24. Iyer JG, Afanasiev OK, McClurkan C, Paulson K, Nagase K, Jing L, et al. Merkel cell polyomavirus-specific CD8⁺ and CD4⁺ T-cell responses identified in Merkel cell carcinomas and blood. *Clinical Cancer Research*. 2011;17:6671–80. [PubMed: 21908576]
25. Yin R, Ribeiro-Filho HV, Lin V, Gowthaman R, Cheung M, Pierce BG. TCRmodel2: high-resolution modeling of T cell receptor recognition using deep learning. *Nucleic Acids Res*. 2023;51:W569–76. [PubMed: 37140040]
26. Huang H, Wang C, Rubelt F, Scriba TJ, Davis MM. Analyzing the Mycobacterium tuberculosis immune response by T-cell receptor clustering with GLIPH2 and genome-wide antigen screening. *Nature biotechnology*. 2020;1–14.
27. Goncharov M, Bagaev D, Shcherbinin D, Zvyagin I, Bolotin D, Thomas PG, et al. VDJdb in the pandemic era: a compendium of T cell receptors specific for SARS-CoV-2. *Nat Methods*. 2022;19:1017–9. [PubMed: 35970936]
28. Tickotsky N, Sagiv T, Prilusky J, Shifrut E, Friedman N. McPAS-TCR: a manually curated catalogue of pathology-associated T cell receptor sequences. *Bioinformatics*. 2017;33:2924–9. [PubMed: 28481982]
29. Zhang W, Wang L, Liu K, Wei X, Yang K, Du W, et al. PIRD: Pan immune repertoire database. *Bioinformatics*. 2019;36:897–903.
30. Mimitou EP, Cheng A, Montalbano A, Hao S, Stoeckius M, Legut M, et al. Multiplexed detection of proteins, transcriptomes, clonotypes and CRISPR perturbations in single cells. *Nature Methods*. 2019;1–9. [PubMed: 30573832]
31. Daniels J, Doukas PG, Escala MEM, Ringbloom KG, Shih DJH, Yang J, et al. Cellular origins and genetic landscape of cutaneous gamma delta T cell lymphomas. *Nat Commun*. 2020;11:1806. [PubMed: 32286303]
32. Gherardin NA, Waldeck K, Caneborg A, Martelotto LG, Balachander S, Zethoven M, et al. $\gamma\delta$ T cells in Merkel cell carcinomas have a proinflammatory profile prognostic of patient survival. *Cancer Immunol Res*. 2021;canimm.0817.2020.
33. Davey MS, Willcox CR, Hunter S, Kasatskaya SA, Remmerswaal EBM, Salim M, et al. The human V δ 2+ T-cell compartment comprises distinct innate-like V γ 9+ and adaptive V γ 9- subsets. *Nat Commun*. 2018;9:1760. [PubMed: 29720665]
34. Creed JH, Wilson CM, Soupier AC, Colin-Leitzinger CM, Kimmel GJ, Ospina OE, et al. spatialTIME and iTIME: R package and Shiny application for visualization and analysis of immunofluorescence data. *Bioinformatics*. 2021;37:4584–6. [PubMed: 34734969]
35. Jin S, Guerrero-Juarez CF, Zhang L, Chang I, Ramos R, Kuan C-H, et al. Inference and analysis of cell-cell communication using CellChat. *Nat Commun*. 2021;12:1088. [PubMed: 33597522]

36. Bolotin DA, Poslavsky S, Mitrophanov I, Shugay M, Mamedov IZ, Putintseva EV, et al. MiXCR: software for comprehensive adaptive immunity profiling. *Nat Methods*. 2015;12:380–1. [PubMed: 25924071]
37. Li H, van der Leun AM, Yofe I, Lubling Y, Gelbard-Solodkin D, van Akkooi ACJ, et al. Dysfunctional CD8 T Cells Form a Proliferative, Dynamically Regulated Compartment within Human Melanoma. *Cell*. 2019;176:775–789.e18. [PubMed: 30595452]
38. Pulliam T, Jani S, Jing L, Ryu H, Jojic A, Shasha C, et al. Circulating cancer-specific CD8 T cell frequency is associated with response to PD-1 blockade in Merkel cell carcinoma. *Cell Rep Med*. 2024;101412.
39. Weeden CE, Gayevskiy V, Marceaux C, Batey D, Tan T, Yokote K, et al. Early immune pressure initiated by tissue-resident memory T cells sculpts tumor evolution in non-small cell lung cancer. *Cancer Cell*. 2023;41:837–852.e6. [PubMed: 37086716]
40. de Vries NL, van de Haar J, Veninga V, Chalabi M, Ijsselstein ME, van der Ploeg M, et al. $\gamma\delta$ T cells are effectors of immunotherapy in cancers with HLA class I defects. *Nature*. 2023;1–8.
41. Wu Y, Kyle-Cezar F, Woolf RT, Naceur-Lombardelli C, Owen J, Biswas D, et al. An innate-like V δ 1+ $\gamma\delta$ T cell compartment in the human breast is associated with remission in triple-negative breast cancer. *Sci Transl Med*. 2019;11:eaax9364.
42. Oshi M, Newman S, Tokumaru Y, Yan L, Matsuyama R, Kalinski P, et al. Plasmacytoid Dendritic Cell (pDC) Infiltration Correlate with Tumor Infiltrating Lymphocytes, Cancer Immunity, and Better Survival in Triple Negative Breast Cancer (TNBC) More Strongly than Conventional Dendritic Cell (cDC). *Cancers*. 2020;12:3342. [PubMed: 33198125]
43. Conrad C, Gregorio J, Wang Y-H, Ito T, Meller S, Hanabuchi S, et al. Plasmacytoid Dendritic Cells Promote Immunosuppression in Ovarian Cancer via ICOS Costimulation of Foxp3+ T-Regulatory Cells. *Cancer Res*. 2012;72:5240–9. [PubMed: 22850422]
44. Boersma B, Jiskoot W, Lowe P, Bourquin C. The interleukin-1 cytokine family members: Role in cancer pathogenesis and potential therapeutic applications in cancer immunotherapy. *Cytokine Growth F R*. 2021;62:1–14.
45. He S, Bhatt R, Brown C, Brown EA, Buhr DL, Chantranuvatana K, et al. High-plex imaging of RNA and proteins at subcellular resolution in fixed tissue by spatial molecular imaging. *Nat Biotechnol*. 2022;40:1794–806. [PubMed: 36203011]
46. Mann SE, Zhou Z, Landry LG, Anderson AM, Alkanani AK, Fischer J, et al. Multiplex T Cell Stimulation Assay Utilizing a T Cell Activation Reporter-Based Detection System. *Front Immunol*. 2020;11:633. [PubMed: 32328071]
47. Heemskerk MHM, Hoogeboom M, de Paus RA, Kester MGD, van der Hoorn MAWG, Goulmy E, et al. Redirection of antileukemic reactivity of peripheral T lymphocytes using gene transfer of minor histocompatibility antigen HA-2-specific T-cell receptor complexes expressing a conserved alpha joining region. *Blood*. 2003;102:3530–40. [PubMed: 12869497]
48. Fleming SJ, Chaffin MD, Arduini A, Akkad A-D, Banks E, Marioni JC, et al. Unsupervised removal of systematic background noise from droplet-based single-cell experiments using CellBender. *Nat Methods*. 2023;20:1323–35. [PubMed: 37550580]
49. Jumper J, Evans R, Pritzel A, Green T, Figurnov M, Ronneberger O, et al. Highly accurate protein structure prediction with AlphaFold. *Nature*. 2021;596:583–9. [PubMed: 34265844]

Statement of Significance

MCC serves as a model of ICB response. We utilized the largest-to-date, multi-modal MCC dataset (n=116 patients) to uncover unique tumor-intrinsic properties and immune circuits that predict response. We identified CD8 Trm and Vδ1 T cells as clinically actionable mediators of ICB response in MHC high and low MCCs, respectively.

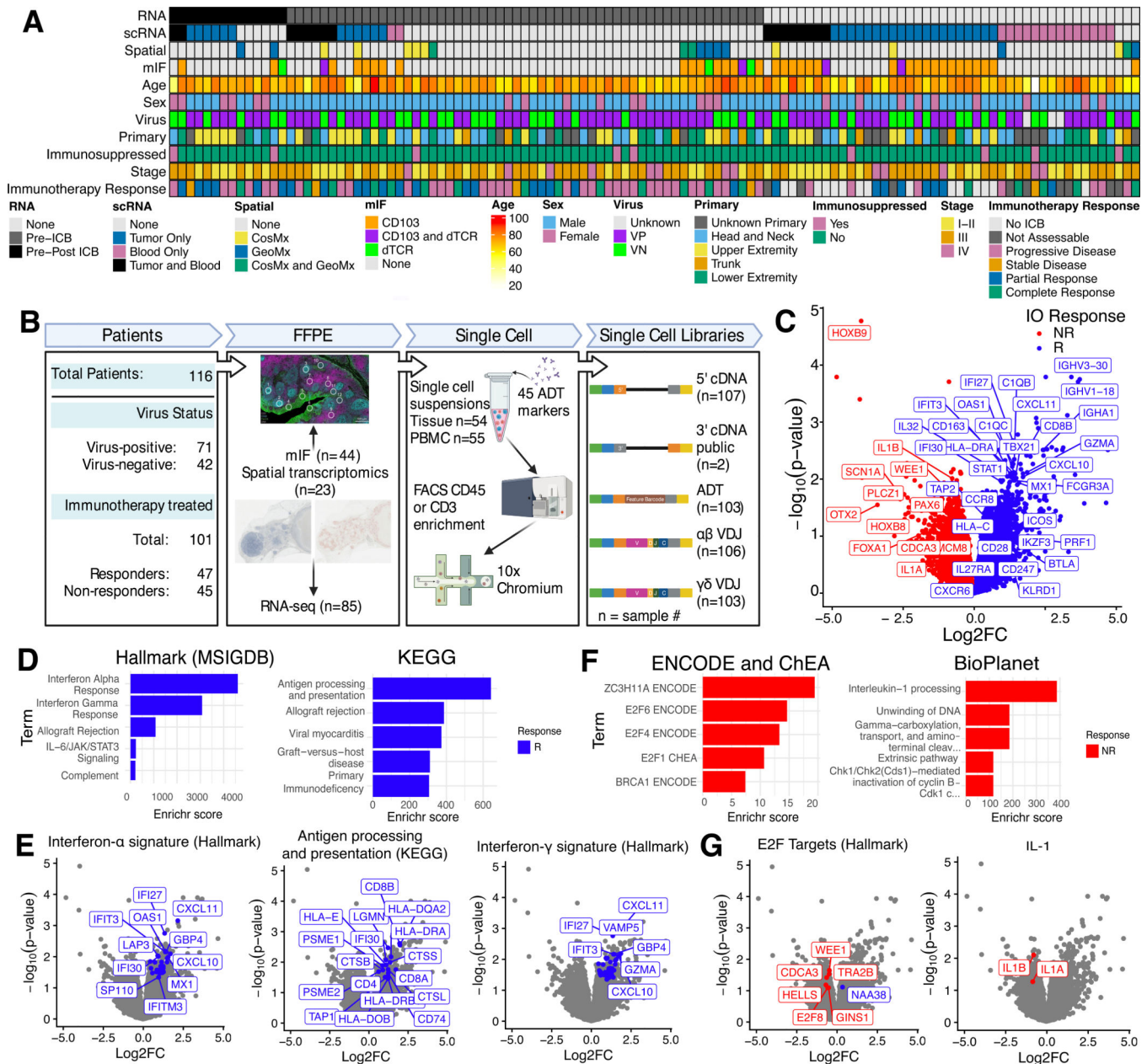


Fig. 1: Differentially expressed genes identify key immune and tumor signatures distinguishing immunotherapy responders and non-responders.

A, Patient sample heatmap of assays and clinical metadata for each sample in our dataset.

Each column represents a single patient with the available assays and associated clinical characteristics. In total, 116 patients were included in this study. 101 patients received immunotherapy. 71 patients were assayed by bulk RNA-seq, 64 by single cell (sc)RNA-seq, 41 by multiplex immunofluorescence (mIF), and 20 by spatial transcriptomics (GeoMx and CosMx). VP, Virus-positive; VN, Virus-negative

B, Schematic of the study overview. Patient characteristics, paraffin-embedded (FFPE) sample sequencing breakdown with sample counts, single-cell processing workflow with sample counts, and single-cell library breakdown with sample counts. Fifty-four tumor

samples and 55 peripheral blood mononuclear cell (PBMC) samples were stained with 45 antibody-derived tags (ADT) and enriched for live, CD45⁺ cells (tissue), or CD3⁺ cells (PBMCs) via fluorescence-activated cell sorting (FACS). Cells were processed using 10x Chromium and four sequencing libraries were prepared for each sample: 5' cDNA, ADT, $\alpha\beta$ VDJ, and $\gamma\delta$ VDJ.

C, Volcano plot of differentially expressed (DE) genes between immunotherapy (IO) responders (R) and non-responders (NR). Bulk RNA-seq was conducted on 63 FFPE tumor samples from 62 immunotherapy naïve, systemic therapy-naïve patients, all samples prior to evaluable responses to ICB. Differentially expressed genes were identified using the DESeq2 package, with the following covariates: sex, viral status, and tissue. Selected protein-coding genes with a false discovery rate (FDR) < 0.1 were labeled. A total of 800 genes were upregulated in responders, and 436 genes were upregulated in non-responders.

D, Pathway analysis of differentially expressed (DE) genes in responders. Only protein-coding genes with FDR < 0.1 were included in the analysis. The top five enriched gene sets from the selected gene set libraries are shown for the responder DE genes. Gene sets were ranked based on the Enrichr combined score, which is a measure of both Fisher's exact test p-value and the z-score of the deviation from the expected rank of the gene set.

E, Three gene sets enriched in responders with associated genes are represented on the volcano plot of DE genes from responders and non-responders. Genes with FDR < 0.1 are labeled.

F, Enrichment gene list analysis of DE genes found in non-responders. Only protein-coding genes with FDR < 0.1 were included in the analysis. The top five enriched gene sets from the selected gene set libraries are shown for non-responder DE genes. Gene sets were ranked based on the Enrichr combined score.

G, Two gene sets enriched in non-responders with associated genes are represented on the volcano plot of DE genes from responders and non-responders. Only genes with FDR < 0.1 are labeled.

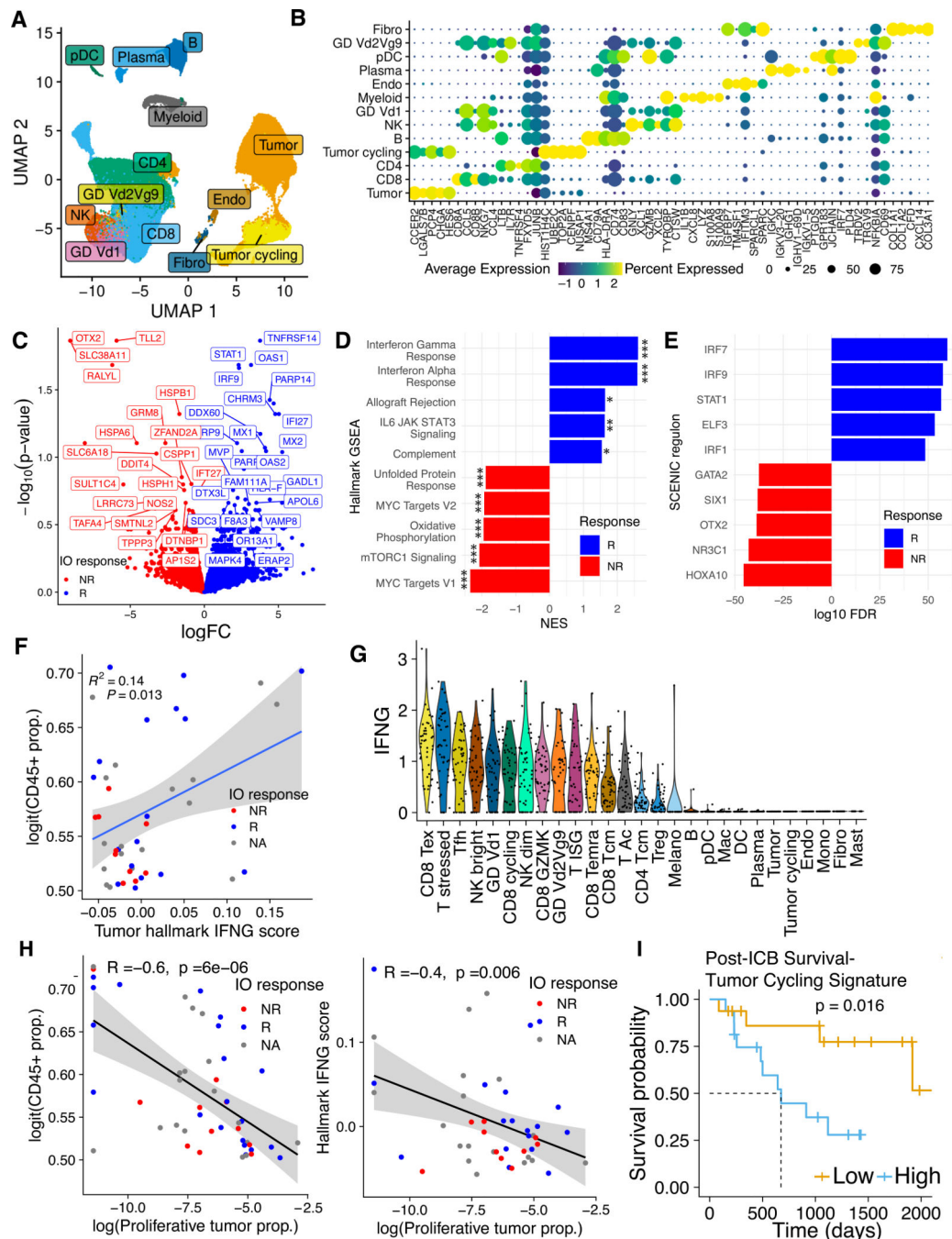


Fig. 2: Response to ICB is associated with tumor cell upregulation of an interferon- γ signature and negatively correlated with a tumor cycling signature.

A, Uniform Manifold Approximation and Projection (UMAP) projection of 139,845 cells among all 49 tissue samples in single-cell (sc) transcriptome datasets with major cell-type annotation. PCA inputs were batch-corrected using the Harmony package and cells were clustered using the standard Seurat pipeline.

B, Dot plot showing the scaled expression of the top five marker genes for each cell type annotated in the tissue scRNA-seq dataset.

C, Differential gene expression analysis of 60,902 tumor cells pseudobulked by tissue sample between ICB responders and non-responders was performed using the edgeR LRT test with sex and virus status as a covariate. 34 genes were upregulated in responders, and 25 genes were upregulated in non-responders. Protein-coding genes with FDR <0.25 are labeled.

D, Gene set enrichment analysis (GSEA) of DE genes found in immunotherapy responders (R) and non-responders (NR). Hallmark gene sets were ranked by normalized enrichment score (NES), with the top 5 gene sets from responders and non-responders listed. **** p adjust<0.001, ** p adjust< 0.01, * p adjust< 0.05

E, SCENIC regulon analysis was run on 60,902 tumor cells. The top 5 regulons differentially expressed in responders (R) and non-responders (NR) are shown, ranked by false discovery rate (FDR), Wilcoxon test.

F, Scatter plot of CD45+ infiltration (% CD45+ cells from FACS) with logit transformation versus Hallmark Interferon- γ signature module score in tumor cells from scRNA-seq, which depicts a positive Pearson correlation between interferon- γ response in the tumor and overall immune infiltration. Samples are colored according to ICB response. *IFNG* score was calculated on tumor cells by the ModuleScore function in Seurat using the MSIGDB “Hallmark Interferon Gamma Response” genes as input.

G, A violin plot of *IFNG* expression of each cell type split by samples and ranked by mean expression.

H, Scatter plots of immune proportion (% CD45+ cells from FACS) with logit transformation or tumor cell *IFNG* score compared to the proportion of proliferating tumor cells among the total cells with log transformation in each scRNA-seq tissue sample. Samples are colored by ICB response, which shows a negative Pearson correlation between the proportion of proliferating tumor cells and either the immune infiltration or *IFNG* response in the tumor cells, with non-responders clustered in the high proliferation, low *IFNG* or low CD45+ % quartile.

I, A Kaplan-Meier curve of post-immune checkpoint blockade (ICB) survival shows that samples with higher levels of tumor cycling cell signature (derived from scRNA-seq) had lower survival. Samples from the bulk RNA-seq dataset were scored based on the “tumor cycling” cell type signature derived from our single cell data with gene set variance analysis (GSVA) and split into the top or bottom quartile (see Methods). Statistical test: Cox proportional hazards.

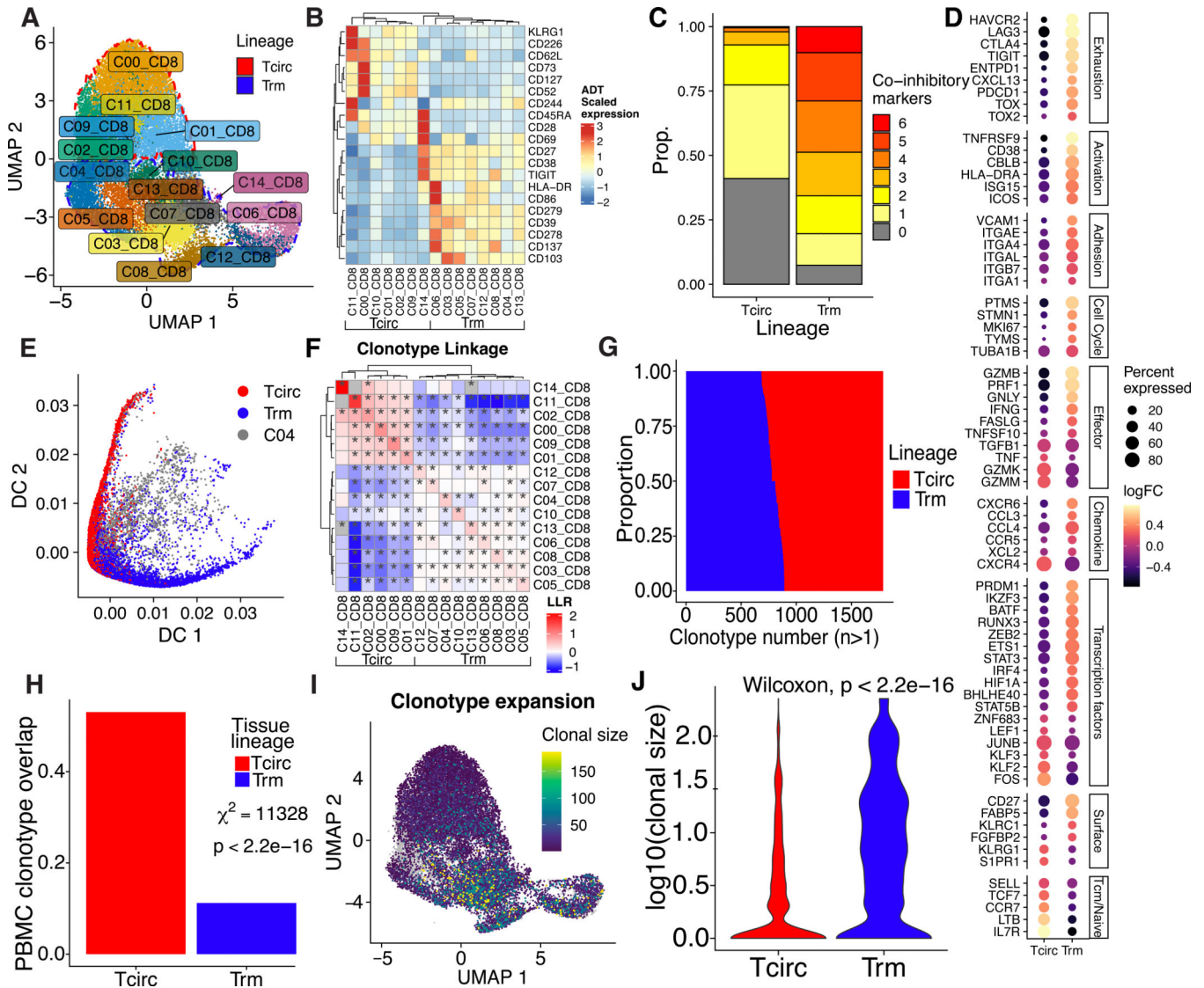


Fig. 3: CD8 TILs form two distinct developmental trajectories composed of circulatory T cells and Tissue resident memory subsets.

A, Uniform Manifold Approximation and Projection (UMAP) projections of 24,448 CD8 T cells in 15 cell clusters. Outlines of the clusters show the formation of two meta-clusters: circulating T cells (Tcirc) and tissue-resident memory (Trm) T cells. Cells were subset from all tissue scRNA-seq samples and clustered using the standard Seurat pipeline.

B, Heatmap of the antibody-derived tag (ADT) protein average scaled expression of each CD8 cluster with hierarchical clustering into two major meta-clusters distinguishing Tcirc and Trm cells.

C, Bar chart of the proportion of co-expressing, co-inhibitory markers statistically enriched on T resident (Trm) compared to T circulatory (Tcirc) cells (χ^2 , $p < 2.2e-16$).

D, Dot plot of selected average marker gene expression for each CD8 lineage with functionally grouped genes. logFC: log2 fold change

E, Diffusion mapping projection of CD8 T cells with cell cycle genes regressed out using the Destiny package. The diffusion map shows two separate lineages, which are colored by

Trm or Tcirc. Cells in the CD8 cluster C04 are depicted in gray. C04 show characteristics of both Trm and Tcirc cells. DC: diffusion component

F, Heatmap of the log-likelihood ratio (LLR) of shared TCR clones between CD8 clusters with the heatmap clustered by LLR demonstrates the formation of two meta-clusters that match the Trm and Tcirc lineages. *, $p < 0.05$, Fisher's test

G, Bar chart of all expanded CD8 clonotypes in the tissue dataset. Each line represents the proportion of cells in each lineage of the single clonotype. The clonotypes were ranked from the highest proportion of cells in the Trm lineage to the lowest proportion (highest in the Tcirc lineage). For 88% of the clonotypes, 100% of the cells were restricted to either the Trm or Tcirc lineage.

H, Bar chart of the proportion of clones that were also found in the PBMC single-cell dataset split by developmental lineage. Only matched blood samples were included in the analysis. Chi-square test for significant differences between proportions.

I, UMAP projection (same as panel A) colored by the TCR clone size.

J, Violin plots of clonal size of all clonotypes found in the Trm or Tcirc lineage.

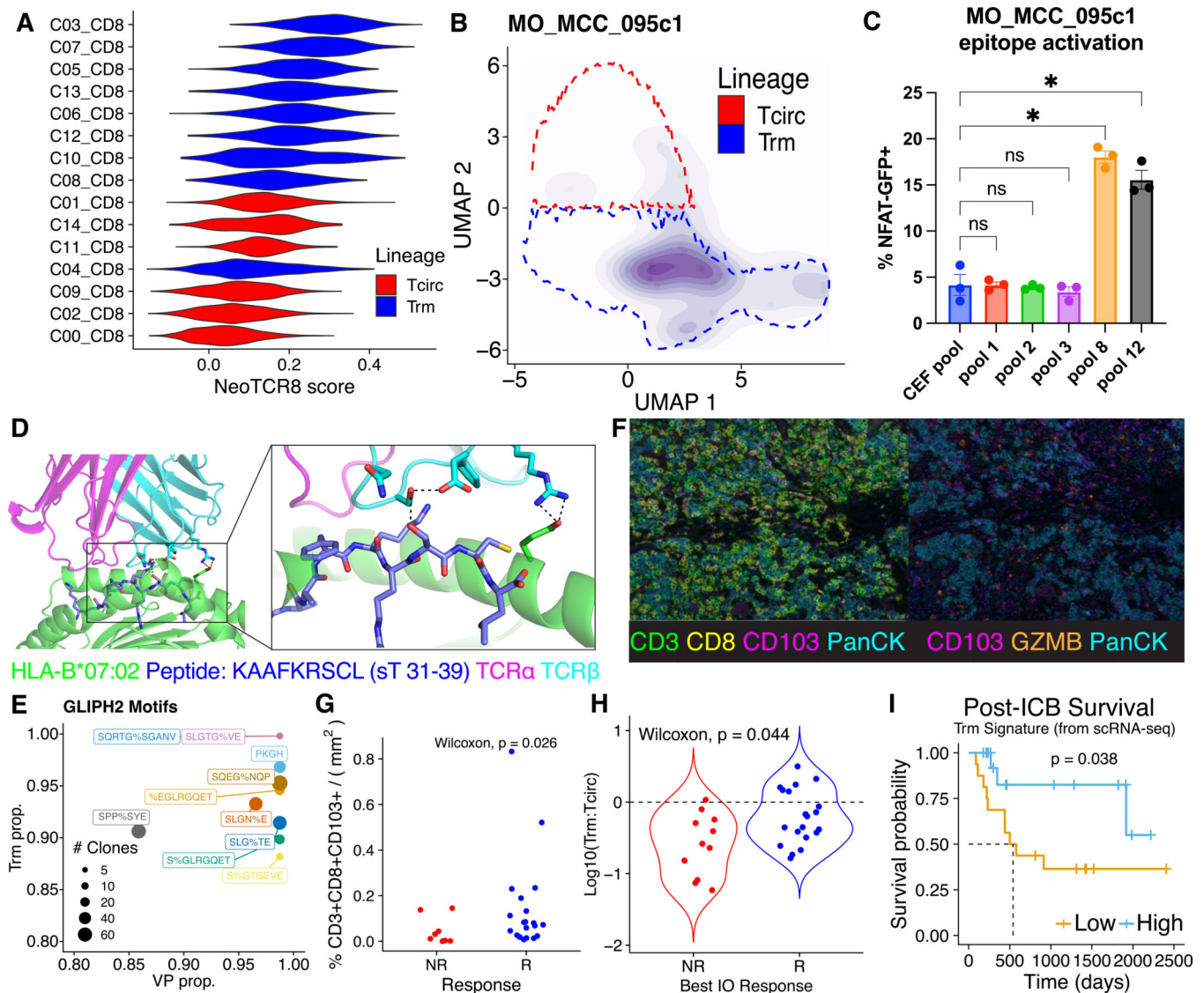


Fig. 4: CD8 Trm cells are tumor specific and prognostic of ICB response.

A, Violin plot of cellular expression of the NeoTCR8(23) module score split by CD8 cluster. Clusters are colored by developmental lineage: T circulatory cells (Tcirc) or T resident cells (Trm).

B, Cell density plot showing the distribution of cells with the MO_MCC_095c1 TCR clonotype, a highly expanded Trm TCR clone found in virus-positive MCC samples. Outline shows the distribution of Tcirc and Trm cells.

C, Jurkat-76 cells modified with an NFAT-GFP reporter and CD8 α /CD8 β were transduced and selected to express the MO_MCC_095c1 TCR. These cells were co-cultured with an autologous lymphoblastoid cell line loaded with pooled peptides from the small and large T antigens of Merkel cell polyoma virus. After co-culture, GFP expression on the reporter cells was measured as a measure of antigen recognition, leading to T cell activation. CEF (CMV, EBV, Influenza) peptide pool, pool 1, and pool 2, pool 3 are representative examples

of negative hits, whereas pools 8 and 12 are positive hits. See Supplementary Fig. S14 shows the complete results for all pools. *, $p < 0.05$, One-way ANOVA.

D, Structural model of the MO_MCC_095c1 TCR in complex with its target peptide, small T (sT) antigen residues 31–39, bound to its putative MHC allele, HLA B*07:02, reveals a canonical TCR-pMHC binding mode and multiple polar contacts that stabilize the interaction (dashed lines). TCR α is magenta, TCR β cyan, peptide slate (stick representation), and MHC green. The high confidence of the model by AlphaFold score (0.82 out of 1) indicates a high likelihood of structural accuracy.

E, Scatter plot of all significant GLIPH2(26) complementary-determining region 3 (CDR3) motifs found in the CD8 tissue TCR dataset. Axis has been adjusted to show motifs that are >80% found in virus-positive samples and >80% found in Trm cells. These virus-positive (VP) Trm motifs are shared between multiple patients and are highly expanded and represent likely shared anti-MCC TCR motifs. “%” indicates a variable amino acid found in the GLIPH motif.

F, Representative multiplex immunofluorescence (mIF) images show skin resident CD8 T cells (CD3+CD8+CD103+) infiltrating the tumor parenchyma, marked by pan-cytokeratin (PanCK), and are cytotoxic, marked by granzyme B (GZMB). Scale bar:100 μ m.

G, A scatter plot of tissue-resident CD8 T cell densities (percentage of CD3+CD8+ expressing CD103 per mm^2) from 29 mIF samples from 26 patients (17 responders, 9 non-responders). Wilcoxon test.

H, A violin plot of the ratio of Trm to Tcirc cells split by response to immunotherapy (IO). Wilcoxon test.

I, Kaplan-Meier curve of survival from the start of ICB from the bulk RNA-seq dataset. Samples were split based on the top and bottom quartiles of the Trm gene set variation analysis score derived from our single-cell dataset (see Methods section). Statistical test: Cox proportional hazards.



Author Manuscript

C, Differentially expressed genes (pseudobulked, edgeR LRT test) in $\gamma\delta$ /NK cells compared to samples with low tumor HLA I expression to samples with high tumor HLA I expression. **D**, A chord diagram of γ and δ V chain pairings for all cells in the tissue scRNA-seq dataset. **E**, Dot plot of average expression of selected significant, Wilcoxon gene markers of $\gamma\delta$ and NK cells in both the tissue datasets separated by developmentally distinct δ -chain or NK lineage. δ 1 and δ 3 T cells show higher levels of exhaustion markers, activation markers, killer cell immunoglobulin-like receptor (KIR) family members, cytotoxic effectors, *CXCR6*, adhesion molecules, and exhaustion-related transcription factors (TF). δ 2 T cells express high levels of TNF, LTB, naïve/T central memory (Tcm) markers, and AP-1 family members (*FOS*).

F, Violin plot of selected significant Wilcoxon protein markers separated by tissue/peripheral blood mononuclear cell (PBMC) samples and grouped by δ -chain or NK lineage.

G, A volcano plot of differentially expressed protein markers (ADT) of V δ 1 cells versus V δ 2 cells. V δ 1 cells upregulated exhaustion and activation markers. V δ 2 cells upregulated naïve and T central memory markers. All labeled proteins had $P_{adj.} < 0.05$. Wilcoxon test.

H, Multiple alignment and consensus sequence of δ CDR3 of the top expanded V δ 1 clones shows clonal focusing with sequence similarity between patients. See Methods for details on TCR selection.

I, UMAP projection of V δ 1 cells (same projection as panel b) colored by pseudotime as calculated by Monocle demonstrate that “root” cells are found mainly in C08_Vd1 whereas differentiated cells are found in C02_Vd1. Root cells were selected based on their highest expression of naïve-like markers, including *TCF7*, CD45RA, CD127, and CCR7.

J, Heatmap of genes that are significantly and non-linearly correlated with V δ 1 pseudotime. V δ 1 T cells have high expression of naïve markers like *TCF7* as well as AP-1 family members at the beginning of pseudotime. During differentiation, V δ 1 cells upregulate co-inhibitory, cytotoxic, activating, and Trm makers. Genes are colored by scaled expression across pseudotime.

K, Representative image of multiplex immunofluorescence (mIF) showing cytotoxic granzyme B (GZMB+) $\gamma\delta$ T cells δ TCR (dTCR+) infiltrating the tumor parenchyma, marked by pan-cytokeratin (PanCK), as indicated by arrows. Scale bar: 50 microns.

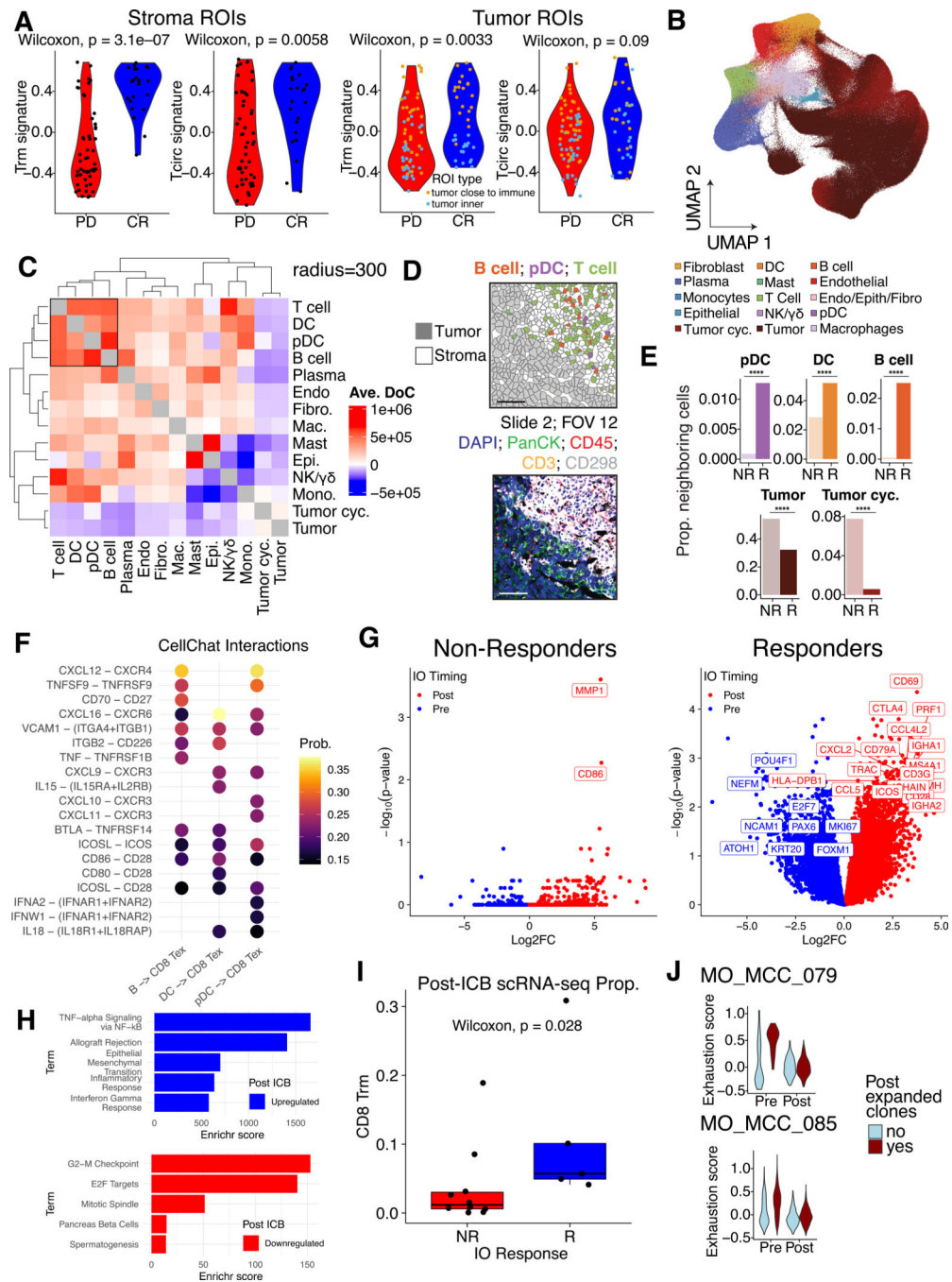


Fig. 6: Spatial and temporal analysis of tumor infiltrating lymphocytes

A, Violin plots of the relative expression of GeoMx regions of interest (ROI) of Tissue resident memory (Trm)/circulatory T (Tcirc) signature genes (derived from our scRNA-seq dataset) by module score in the stroma or tumor parenchyma.

B, UMAP projection of 614,209 cells segmented using the CosMx assay. The cells are colored according to the inferred cell type. Cells were clustered in Seurat and batch-corrected using SCTransform.

C, Unsupervised hierarchical clustering heatmap of cell type co-localization as measured by the degree of co-localization (DoC) within CosMx fields of view (FOV). Neighboring cells were identified within a 54 micron radius of each cell, and then the (DoC) was determined using the SpatialTIME(34) method. A high degree of co-localization was observed among T cells, B cells, DCs, and pDCs, is outlined in the upper left. Ave: average; DoC: degree of co-localization. Mac: Macrophages, Endo: endothelial cells, Fibro: fibroblasts, Tumor cyc: tumor cycling cells, pDC: plasmacytoid dendritic cells, DC: dendritic cells, Epi: epithelial cells, Mono: monocytes

D, A sample FOV showing co-localization of T cells, B cells, and pDC at the tumor-stroma interface using CosMx cell segmentation representation (left) and morphological markers showing para-nuclear dot-like expression of Pan cytokeratins (CK), with high CD298 expression across immune cells. Scale bar: 50 μ m.

E, Bar charts of the cell types found in T cell neighborhoods (54 μ m radius) showing prevalence of pDC, DC, and B cells in T cell neighborhoods of responders (R) versus T cell neighborhoods of non-responders (NR). Comparisons were made using the chi-square test, ****, p value < 2.2e-16.

F, A dot plot of receptor-ligand interactions from CellChat on the scRNA-seq dataset identifies multiple activating chemokines, cytokines, and surface receptor interactions between B cells, DCs, and pDCs to exhaustion, CD8 Trm cells (CD8 Tex.).

G, Differential gene expression analysis (DE-seq2) comparing matched pre- versus post-ICB sample pairs across 7 non-responders and 7 responders shows upregulation of inflammatory genes after ICB treatment. Among responders, 1560 genes were upregulated and 995 genes were downregulated after therapy. (DE-Seq2, $P_{adj.} < 0.1$.) Only two differentially expressed genes were identified among the non-responders.

H, Enrichr gene list analysis of DE genes in pre- versus post-ICB matched pairs from responders shows inflammatory pathways upregulated and cell cycle pathways downregulated after treatment. Only protein-coding genes with FDR < 0.1 were included in the analysis. The top five enriched gene sets from the Hallmark gene set libraries are shown for the DE genes. Gene sets were ranked based on the Enrichr combined score, which is a measure of both Fisher's exact test p-value and the z-score of the deviation from the expected rank of the gene set.

I, Corrected cellular proportions of Trm cells in 15 post-ICB treated samples show CD8 Trm infiltration is higher (3.01-fold) in partial responders (PR) versus progressive disease (PD) after treatment. Samples are split by response prior to biopsy. Five of the samples represent a new cohort of post-ICB samples used to verify results from the original scRNA-seq dataset. Wilcoxon test.

J, Violin plots of CD8 cells in MO_MCC_079 and MO_MCC_085, two scRNA-seq pre/post sample pairs. Exhaustion score expression was split by treatment timeline and grouped by clonotype expansion showed that expanded cells were more likely to be exhausted pre-treatment, with exhaustion levels significantly decreased after treatment. The exhaustion score(37) was derived from Seurat AddModuleScore.

Table 1.

Clinical characteristics of patients included in this study with odds ratio of how characteristics affect immunotherapy response.

Characteristics	CR	PR	Total Responders	SD	PD	Total Non-Responders	NA	No ICB	Pvalue	OR	95% CI
Gender											
Male	15	24	39	5	30	35	8	10	0.54	1.39	0.49–3.92
Female (ref)	4	4	8	0	10	10	1	5	1.00	1.00	
Age											
>=65	15	19	34	4	29	33	8	12	0.92	0.95	0.38–2.38
<65 (ref)	4	9	13	1	11	12	1	3	1.00	1.00	
Stage at Diagnosis											
Stage III	13	14	27	3	22	25	4	7	0.99	1.00	0.40–2.54
Stage IV	1	5	6	1	6	7	0	0	0.75	0.80	0.21–3.00
Stage I-II (ref)	5	9	14	1	12	13	5	8	1.00	1.00	
Immunocompromised											
yes	0	1	1	1	5	6	1	1	0.05	0.14	0.02–1.22
no (ref)	19	27	46	4	35	39	8	14	1.00	1.00	
Viral Status											
Positive	13	13	26	3	27	30	5	10	0.33	0.65	0.28–1.52
Negative (ref)	6	14	20	2	13	15	2	5	1.00	1.00	
Type of ICB											
CTLA-4/PD-1	2	10	12	1	0	1	1	0	0.01	10.76	1.31–88.52
PD-L1	2	4	6	2	16	18	2	0	0.02	0.30	0.10–0.87
PD-1 (ref)	15	14	29	2	24	26	6	0	1.00	1.00	
Radiation Naïve											
yes	7	13	20	2	20	22	6	0	0.55	0.77	0.34–1.76
no (ref)	12	15	27	3	20	23	3	0	1.00	1.00	
Chemotherapy Naïve											
yes	15	23	38	3	31	34	9	0	0.55	1.37	0.51–3.69
no (ref)	4	5	9	2	9	11	0	0	1.00	1.00	
ICB Naïve											
yes	19	25	44	3	35	38	9	0	0.18	2.70	0.65–11.18
no (ref)	0	3	3	2	5	7	0	0	1.00	1.00	
Known Primary											
yes	13	25	38	3	37	40	8	13	0.30	0.53	0.16–1.72
no (ref)	6	3	9	2	3	5	1	2	1.00	1.00	
Location of Primary											

Characteristics	CR	PR	Total Responders	SD	PD	Total Non- Responders	NA	No ICB	Pvalue	OR	95% CI
Upper Extremity	7	7	14	0	10	10	4	4	0.58	1.40	0.45–4.38
Trunk	1	1	2	0	5	5	0	2	0.36	0.4	0.06–2.48
Lower Extremity	3	6	9	3	10	13	1	3	0.55	0.69	0.22–2.22
Unknown	6	4	10	2	3	5	1	2	0.33	2.00	0.52–7.63
Head & Neck (ref)	2	10	12	0	12	12	3	4	1.00	1.00	

CR, complete response; PR, partial response; SD, stable disease; PD, progressive disease; OR, odds ratio; NA, not applicable; ICB, immune checkpoint blockade; CI, confidence interval, ref, reference

Author Manuscript

Author Manuscript

Author Manuscript

Author Manuscript